

Response-Induced Geometry of the Admissibility Substrate

Quotient Response Geometry, Intrinsic Cost, and the Weak-Field Trace Obstruction

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Abstract

This paper develops the geometric-substrate program of the Admissibility Physics Framework as a response-geometry program. Its upstream object is now the structurally aligned held continuation profile of Paper 10: a complete pre-record carrier whose contextual identity is preserved through admissible coherent continuation. The central type distinction is among the held-profile carrier V_H , its represented process algebra $\mathfrak{T}_H$, a finite response-coordinate space and response-null quotient, the invariant committed-record image $\mathbf{R}(H)$ and the outcome-bearing collapse maps $\{\mathbf{C}_i\}$, and the physical comparison substrate. The represented process algebra supplies the operational target of the response map and keeps held continuation geometry, response geometry, record formation, and substrate geometry distinct. Once redundant response coordinates are quotiented, a differentiable response map pulls the operational process norm back to a positive homogeneous tangent gauge

$$F_{\text{resp}}(q, v) = \|D\bar{\Phi}_q(v)\|_{\text{proc}}.$$

This gauge defines an intrinsic response length and path distance. In a finite linear chart it is flat and translation invariant, and the resulting distance theorem is exactly the geometric form of the finite quotient, upper-gain, minimum-gain, and bi-Lipschitz package developed in Paper 2.

The response distance is not automatically the physical interaction ledger. This paper proves the precise calibration theorem: if the physical ledger rate is bounded above and below by the pullback gauge, then the two induced path costs are bi-Lipschitz equivalent, with equality only under exact rate calibration. It also separates one-homogeneous path cost from the two-homogeneous residual energy of smooth residual realization. In the latter branch the local theorem is

$$I_{\text{H}}^{\text{res}}(q) = \frac{1}{2}d_{g^{\text{E}}}(q_0, q)^2 + o(d_{g^{\text{E}}}^2),$$

not an identification of interaction energy with distance itself.

The intrinsic response-orbit quotient is constructed kernel-first, but no manifold and no physical substrate anchor are assumed upstream. Finite-stage executable tests are organized into precision-refinement channels: each stage has a finite outcome set, while the compatible tower may carry a real-valued limit. A conditional finite operational-basis theorem controls operational distinction content; a separate faithful-realization gate is required before that basis becomes a jointly realizable, state-separating precision-channel family. Quotient-faithful channel limits generate the record topology, while second countability additionally requires a countable specification grammar and a countable topology-generating subfamily. Lossless relabelling preserves the physical quotient and smooth saturation supplies the minimal Sikorski functional calculus. The resulting object is a differential space with derivation tangent

$$T_p^{\text{der}}P = \text{Der}_p(\mathcal{C}_p, \mathbb{R}),$$

canonically dual to $\mathfrak{m}_p/\mathfrak{m}_p^2$. Finite smooth generation is isolated as a separate subcartesian gate; on that locus the established orbit theorem makes complete-family continuation orbits immersed smooth manifolds. Rank-changing, non-flowable, or non-finitely-presented sectors remain intrinsic singular strata.

On a smooth continuation orbit, set-level closed-world fibre completeness is not enough for first-order faithfulness. If the complete response and a finite local APF readout generator factor smoothly through one another, then

$$\ker D\beta_x = N_x^{\text{ctx}} = \bigcap_{a \in \mathcal{A}_x} \ker \ell_{a,x},$$

and the induced quotient response is injective. Its process norm then defines an intrinsic positive response-orbit gauge. Transport of that gauge to physical site displacements remains a separate substrate-anchor gate. Quadraticity may convert the response gauge into a positive cost metric, while Lorentzian comparison geometry still requires an independent causal cone and signed two-homogeneous structure.

The weak-field program is retained and sharpened. In each clock–ruler comparison plane the static load response has generator L . If the loaded comparison kernel is $q_\chi = R_\chi^* q_0$, then

$$\left. \frac{d}{d\chi} \log \sqrt{|\det q_\chi|} \right|_{\chi=0} = \text{tr } L.$$

After clock calibration this scalar volume mode is exactly $1 - \gamma_C$. The remaining APF-specific theorem target is therefore the operator statement

$$P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0,$$

where $\mathcal{K}$ is the gauge-fixed response Hessian and $\mathcal{J}$ the scalar recruitment source. It states that static scalar load excites only trace-free exchange and no independent comparison-volume mode. Paper 6 remains the source of record for the represented smooth GR closure; Paper 9 now supplies the intrinsic response geometry beneath that closure and identifies its weak-field scalar obstruction.

Companion technical supplement and closure policy

This main paper is accompanied by *Paper 9 Technical Supplement: Dependency-Closed Conditional Package for Response-Induced Geometry of the Admissibility Substrate*, Version 1.1. The supplement reproduces the premises used by the certified native core and keeps finite joint extension, compatible joint realization, operational-to-channel realization, countable topology generation, finite smooth generation, smooth response-factor completeness, the physical substrate anchor, causal signature, no trace, and nonlinear GR as named gates. No external APF reading is required for that certified native core; Paper 6 remains the separately marked external source of record for the represented nonlinear endpoint.

Result provenance classes

Every load-bearing result in this paper belongs to one of three classes. **APF-native** results are proved from locally printed premises and any explicitly named APF regime gates. **Standard imports** are classical mathematical theorems stated with their exact hypotheses. **Hybrid** results combine an APF certificate with a standard theorem. Open gates and external endpoints are not promoted into the APF-native class by notation.

Governing claim.

Geometry is the represented intrinsic structure of held-profile response, after contextual identity, null response directions, record formation, ledger calibration, substrate anchoring, quadraticity, and causal signature are kept type-distinct.

$$\begin{aligned}
\mathcal{A}_{\text{fin}} &\longrightarrow (P^{\text{red}}, \tau_{\text{rec}}, \mathcal{C}_{\text{APF}}) \\
&\longrightarrow \text{finite differential presentation} \\
&\longrightarrow \text{continuation orbits} \longrightarrow TS/N^{\text{ctx}} \\
&\longrightarrow F_{\text{resp}} \longrightarrow d_{\text{resp}} \longrightarrow \text{calibrated cost geometry} \\
&\longrightarrow \text{signed causal geometry} \longrightarrow \text{volume response.}
\end{aligned}$$

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1. Orientation: the geometric-substrate problem

Paper 10 supplied a continuation-first language for APF. Its central move was to separate admissible possibility from dynamics:

$$\Gamma \Vdash_C D \rightsquigarrow E$$

states that D can continue as E within capacity bound C . A dynamics, probability law, or variational rule is a later selector or weighting on admissible continuation words.

The same paper proposed a compressed reading of geometry:

Geometry is the regularized cost structure of carrying comparison, localization, and transport continuations.

This paper begins the work required to make that sentence theorem-grade.

The geometric-substrate problem has six type-distinct parts:

1. define comparison continuations and finite response models operationally;
2. quotient response-null coordinates and construct intrinsic response length;

3. calibrate that modeled length to the physical interaction ledger;
4. anchor response directions to tangent directions of the comparison substrate;
5. derive a signed causal comparison geometry without confusing it with the positive cost metric;
6. determine how recruitment load deforms the comparison geometry and align the result with the Paper 6 represented-GR closure.

This revision proves the finite geometric statements in items 2 and the conditional calibration statement in item 3. It formulates item 4 as an explicit anchor gate, corrects the signature logic in item 5, and retains the redshift and scalar trace program as the first worked application of item 6. *Boundary* 1.1 (Status discipline). Throughout this paper, P means proved from stated definitions and assumptions in this paper; C means conditional on an explicit gate or structural commitment; R means recovered in a represented regime; I means imported from another APF paper under its stated hypotheses; and O means open. The present document mixes P, C, R, and I: the quotient and pullback theorems are finite mathematics; ledger realization, anchoring, signature, and the no-trace source are conditional or open; redshift is a conditional recovery; and the Einstein closure is imported from Paper 6.

On operational language

This paper develops geometry as the represented cost structure of finite comparison continuations. The continuation primitive imported from Paper 10 is a static algebraic relation $\Gamma \Vdash_C D \rightsquigarrow E$ on continuation profiles, and the metric-emergence ladder of §8 derives the quadratic metric kernel as a representation *conditional on a series of structural gates closing simultaneously* — not as something the framework’s primitives produce by temporal evolution.

The phrasings “becomes available,” “emerges,” “closes,” and “recovers” that appear throughout this paper are the local readings of conditional structural commitments. Where the prose says “the metric becomes available only after several gates close,” the structural reading is “the metric is a representation conditional on the gates’ joint structural validity.” Where the prose says “admissible chart changes become diffeomorphism-type relabelings in the smooth represented regime,” the structural reading is “in the smooth represented regime, the admissible relabelings are exactly the diffeomorphism-type ones.” The “becomes” is locator-language, not process-language.

Time itself is derived in this corpus: Paper 3 derives the arrow of time as the loss of global admissible coordination, and Paper 6 derives the metric and dimensionality of spacetime. The geometric machinery developed here operates on the static comparison structure (c_x, q_x) at the substrate; what reads as “response,” “recruitment,” or “redshift recovery” is the local reading of conditional structural commitments under temporal slicing. A reader who wants the structural reading can take every “becomes available,” “emerges,” and “recovers” as shorthand for the local reading of the corresponding structural conditional. The technical apparatus developed below — the cost germ c_x , the quadratic kernel q_x , the geometric recruitment functional, the cost-curvature response — is already structural; only the pedagogical wrapping is temporal.

This convention matches Paper 0’s “A note on language” in the Prelude §1 and Paper 0’s Descriptive Reading chapter on temporal language and the eternalist commitment, Paper 1 supplement v8.24’s “On the use of operational language” note in §1, and Paper 10 v1.17’s parallel note in §1. The framework is in this respect in the company of block-universe readings of general relativity, the Wheeler–DeWitt timeless quantum gravity programme, and the Page–Wootters mechanism for time

emergence in entangled quantum systems — structural at the foundation, with time as a derived feature of how the structure is read locally.

A note on two “substrate” senses. The framework uses “substrate” in two distinct but related senses, and a reader of this paper alongside Paper 0 should keep them apart. Paper 9 uses *substrate* in its **material-substrate** sense throughout: the *geometric substrate* is the physical object that carries comparison continuations; the *comparison substrate* is the physical object equipped with admissible comparison continuations between sites; *substrate degrees of freedom* are the local mode-content that the recruitment profile $\varphi_d(x)$ recruits to carry a continuation. Paper 0 §sec:substrate_position_axis introduces a different sense: the *substrate-position axis* is a position locus on the framework’s three-regime architecture (void regime / middle regime / saturation regime). The two senses are parallel but not identical: the geometric substrate is *what carries* the comparison continuation; the substrate-position axis is *where in regime space* the architecture sits. Paper 9’s content is largely middle-regime physics in the substrate-position-axis sense (the metric-emergence ladder of §8 requires middle-regime conditions: locally smooth and locally additive comparison cost on a connected comparison substrate without saturation); the saturation-direction reading is articulated in §“Cost-curvature response” where recruitment-load already committed reduces residual stiffness for additional comparison continuations. Paper 6 main v3.9 + Supplement v2.10 carry the substrate-position-axis architectural reading at the dynamics/geometry/cosmology level.

2. Canonical type ledger and corpus dependencies

Paper 9 consumes the mature continuation architecture rather than redefining it. The canonical objects are:

Object	Meaning	Source of record
$H \in \text{Held}$	Structurally aligned, occupied, pre-record continuation profile	Papers 1, 10, and 14
V_H	Complete finite represented carrier of held continuation profiles	Papers 1 and 10; Paper 5 supplies the quantum-sector completion
$\mathfrak{T}_H$	Represented algebra of admissible transformations of held profiles	Papers 1 and 5
$\mathcal{A}_H$	Observable/effect algebra paired with held states and processes	Paper 5
$R(H) \in \text{InvRec}$	Invariant phase-insensitive committed-record image	Papers 10 and 37
$\{\mathbf{C}_i(H)\}$	Outcome-bearing committed records	Papers 5 and 37
$R_\Pi/\sim$	Response coordinates modulo process-null implementation directions	Paper 9
$\mathcal{M}$	Physical comparison substrate carrying localization and transport continuations	Paper 9

The geometry pipeline of this paper is therefore

$$\begin{aligned} H \in \mathbf{Held} &\longrightarrow V_H \longrightarrow \mathfrak{T}_H, \\ T\mathcal{M} &\longrightarrow R_{\Pi}/\sim \longrightarrow \mathfrak{T}_H. \end{aligned}$$

with the final arrow supplied only by a physical comparison-response anchor. Record formation $Q : \mathbf{Held} \rightarrow \mathbf{Rec}$ is downstream and is not used to define the pre-record response geometry.

Audit Note 2.1 (Three distinct planes and three distinct orientations). The elementary quantum comparison carrier of Paper 5, the clock-ruler load-response plane of this paper, and the tangent plane of a Lorentzian comparison substrate are not automatically identical:

$$W_A^{\text{quantum}} \neq W_e^{\text{clock-ruler}} \neq T_x\mathcal{M}$$

unless an explicit anchor theorem identifies them. Likewise distinguish

$$\begin{aligned} o_H &= \text{held complex orientation,} \\ o_{\text{rec}} &= \text{irreversible record order,} \\ o_{\text{caus}} &= \text{time orientation of the causal cone.} \end{aligned}$$

Paper 5 derives o_H in its declared coherent quantum regime. Papers 3 and 37 supply the irreversible record order o_{rec} . Paper 9 may represent that order by o_{caus} only after the causal-cone and signed-signature gates close. No one of these orientations is substituted for another by notation.

Audit Note 2.2 (Cross-paper dependency boundary). Paper 10 supplies continuation grammar, held-profile identity, and the universal held continuation lift. Paper 5 supplies the quantum-sector represented carrier, local coherent circle, and natural complex orientation. Paper 9 neither rederives those quantum results nor uses them to close the substrate anchor, the Lorentzian signature gate, or the weak-field no-trace operator. Paper 6 remains the source of record for the represented nonlinear Einstein closure.

3. Main results and non-claims of this draft

The paper now has two linked theorem spines. The first is intrinsic response geometry; the second is the weak-field deformation of the represented comparison kernel. The table records what is proved, what is conditional, and what remains open.

Result	Status	Meaning
Finite operational basis	Cond./model	General basis existence is conditional on the joint-realizability package; APF-Engine certifies the exact finite atom-cover instantiation and one compatible minimum joint carrier.
Finite physical precision-channel generation	G	Requires a faithful realization preserving joint realizability and recoverability, separating reduced states, and admitting quotient-faithful real limits.

Record topology	P G	Real channel limits generate the initial topology and a separating family gives Hausdorffness. Second countability additionally requires a countable grammar and countable topology-generating subfamily.
Differential-space closure	P/D	Localization and smooth saturation define the minimal Sikorski structure compatible with the admitted record algebra.
Finite-channel sufficiency	P/G	Once a complete jointly realizable channel basis exists, additive standing-capacity accounting makes it finite. Existence follows only after the operational-to-channel gate closes.
Finite differential sufficiency	G	Finite state separation does not imply finite smooth generation of the continuation-regular sheaf.
Smooth continuation orbits	I/P	On the finite-differential-presentation locus, the imported subcartesian orbit theorem gives immersed smooth continuation orbits for complete local flow families.
Regular local response quotient	P H	Constant rank gives quotient charts; derivative-kernel faithfulness additionally requires smooth two-way response factorization.
Response-orbit quotient gauge	P H	The injective quotient response defines an intrinsic positive homogeneous gauge on the response orbit after the smooth-factor gate closes.
Physical comparison-substrate anchor	G	Requires a smooth site map and tangent map equal to its quotient differential, with direct-comparison calibration and stable provenance.
Intrinsic response distance	P	Integrating the response-orbit gauge gives a coordinate-invariant path distance and endpoint process bound.
Linear Paper 2 specialization	P	The finite linear branch is the flat quotient $R_{\Pi}/\ker T_{\Pi}$ and recovers the optimal response gains.
Physical-ledger calibration	P H	Pointwise comparison of ledger rate with the response gauge gives bi-Lipschitz comparison; equality requires exact calibration.
Local residual energy-distance law	P SRR	A smooth positive Hessian gives $I_{\Pi}^{\text{res}} = \frac{1}{2}d_{g^E}^2 + o(d^2)$ near equilibrium.
Quadratic positive cost metric	P Q	A parallelogram certificate converts the response gauge into a positive metric; otherwise the Finsler branch remains.
Lorentzian comparison structure	O/C	Requires an independent finite-coordination cone and signed two-homogeneous structure.

Point-source exterior profile	C/P	An assumed isotropic quadratic exterior kernel gives $\chi(r) \sim 1/r$; kernel and calibration remain inputs.
Clock-sector redshift recovery	R	Calibrated load, clock response, and the proper-time dictionary recover the redshift factor conditionally.
Trace-volume theorem	P/C	The first comparison-volume variation is $\text{tr } L$, equal after clock calibration to $1 - \gamma_C$.
No-trace operator reduction	P/C	The local weak-field target is $P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0$; deriving it from recruitment remains open.
Conditional weak-field exterior	C/R	If no trace closes, the first-post-Newtonian Schwarzschild exterior follows conditionally.
Full represented GR	I/C	External Paper 6 endpoint under H1–H3/A9/Lovelock; finite-continuability translation remains open.

Finite capacity and the positive marginal floor do not by themselves turn an arbitrary closed operational ledger into a jointly realizable finite basis. The general basis theorem uses an explicit finite joint-extension package. The Engine closes the finite atom-cover instantiation and separately exposes the general bridge as open. A further faithful realization gate is required before the operational basis becomes a finite physical precision-channel basis.

Audit Note 3.1 (Non-claims). This paper does not identify every physical interaction cost with a distance, does not derive Lorentzian signature from T^*T , does not infer spacetime dimension from a register-count theorem, and does not turn physical divisibility into the mathematical subdivision of a curve. It does not rederive the Einstein equations, the value of G , the dimension $(1+3)$, or the full equivalence principle. Its new mathematical contribution is the typed record-topology–differential-space–orbit construction, the kernel-first response geometry on emergent smooth branches, and the trace–volume reduction of the weak-field spatial obstruction.

4. Alignment with Paper 6: represented-GR source of record

Paper 6 and its Technical Supplement already contain the represented-geometry route to GR. In that paper the sequence is:

$$\begin{aligned}
\text{cost distance} &\Rightarrow g_{\mu\nu} \Rightarrow (-, +, +, +), \, d = 4 \\
&\Rightarrow \text{capacity-gradient curvature} \Rightarrow \text{A9 closure} \\
&\Rightarrow \text{Lovelock uniqueness} \Rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.
\end{aligned}$$

Paper 9 should therefore not duplicate that proof. Its role is to express the same represented-geometry closure in the newer finite-continuability language and to identify what that closure looks like in the weak-field scalar sector. Of the imported hypotheses, H1 (continuum emergence) and H2 (locality of the Δ_{geo} closure) are discharged by the Recruitment-Radius papers (Papers 24 and 25) via continuous anchor profiles and L_{loc} on substrate kernels; the represented-geometry import that remains is H3 together with the A9/Lovelock uniqueness step.

Theorem 4.1 (Paper 6 represented-GR import). *Under the Paper 6 hypotheses H1–H3, together with the A9 closure and Lovelock uniqueness in the represented smooth Lorentzian regime, the*

admissibility-preserving geometric dynamics are governed by

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

This theorem is imported here as a represented-geometry result, not reproved.

Corollary 4.2 (Weak-field shadow of the Paper 6 closure). *If the Paper 6 represented-GR closure is accepted, then in the static weak-field exterior of a point mass the metric representation has the first-post-Newtonian Schwarzschild form*

$$ds^2 = - \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 + \left(1 + \frac{2GM}{rc^2}\right) d\mathbf{x}^2 + O(c^{-4}),$$

and hence the scalar weak-field response of Paper 9 has

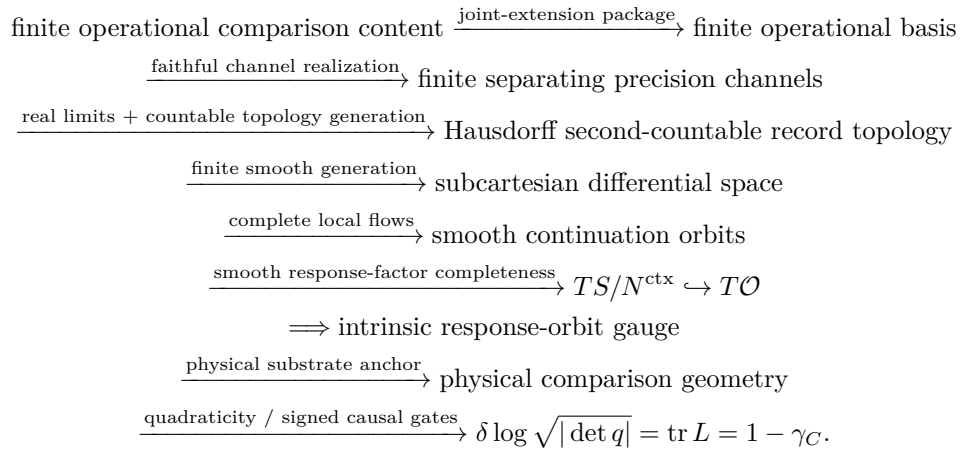
$$\gamma_C = 1, \quad \theta = 1 - \gamma_C = 0.$$

Audit Note 4.3 (What Paper 9 adds beyond importing Paper 6). The import theorem says that the represented smooth geometry satisfies the Einstein equations once the Paper 6 gates close. Paper 9 adds a lower-level translation: redshift fixes the clock response, while the entire leading spatial ambiguity is the trace of a local clock-ruler scalar response generator. Thus the no-trace/unimodular gate is the finite-continuability weak-field face of the Paper 6 A9/Lovelock closure. If Paper 6 is used, $\theta = 0$ follows in the weak-field represented regime. If Paper 9 is developed bottom-up, $\theta = 0$ is the next primitive substrate theorem to prove.

5. Reader path and theorem spine

The paper should be read in two passes. The first pass constructs intrinsic response geometry without presupposing spacetime. The second pass anchors and deforms that geometry in the weak-field comparison sector.

Publication spine.



The decisive closures are therefore:

1. close the faithful operational-to-channel, countable-generation, and finite-smooth-generation gates;
2. certify complete local continuation flows and smooth response-factor completeness on the regular branch;

3. calibrate the intrinsic response-orbit gauge to the physical interaction ledger;
4. construct the physical comparison-substrate anchor and independent causal cone;
5. derive the loaded-equilibrium Hessian $\mathcal{K}$, source $\mathcal{J}$, and the no-trace condition $P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$.

The paper now supplies precise conditional theorems and exposes each remaining bridge separately. Connections and curvature are deferred until regularity, smoothness, strong convexity or quadraticity, the physical anchor, and causal signature have all been certified.

Audit Note 5.1 (Why Paper 9 is the geometric home). Paper 2 supplies a finite response model and its quotient estimates. Paper 9 generalizes those estimates into intrinsic path geometry, anchors them to the comparison substrate, separates positive cost geometry from signed causal geometry, and applies the result to recruitment-deformed clock-ruler response. Paper 2 should therefore import finished Paper 9 theorems later rather than absorb the geometric program itself.

6. Imports from finite continuability

Definition 6.1 (Continuation judgment). *The primitive APF judgment is*

$$\Gamma \Vdash_C D \rightsquigarrow E.$$

It means that, in context Γ , distinction-expression D can continue as distinction-expression E using enforcement capacity no greater than C .

Definition 6.2 (Continuation cost). *The cost of a continuation is*

$$\kappa_\Gamma(D \rightsquigarrow E) := \inf\{C : \Gamma \Vdash_C D \rightsquigarrow E\}.$$

An identity-class continuation is a continuation with $E \in [D]_\Gamma$, where $[D]_\Gamma$ is the tested continuation-equivalence class of D .

Definition 6.3 (Held continuation profile import). *A held continuation profile $H \in \text{Held}$ is the structurally aligned, occupied, pre-record object that carries complete contextual continuation identity together with every consequential coherent distinction still available to later continuation. Its finite represented carrier is V_H . A record-forming class transition is a generally nonfaithful map*

$$Q : \text{Held} \longrightarrow \text{Rec}$$

that commits a tested distinction and may remove globally common orientation while retaining consequential relative information.

Definition 6.4 (Universal held continuation lift import). *For a local comparison coordinate q inside H , Paper 10 defines the minimum complete local carrier*

$$L_q(H) = G_q(H) / \sim_q,$$

the quotient of admissible local continuation germs by complete short-context equivalence. In a smooth operationally first-order branch it is represented by the admitted first-jet locus. Paper 9 uses this only as an identity-complete local carrier; it does not import the Paper 5 quantum S^1 theorem into substrate geometry.

Definition 6.5 (Recruitment profile). *For a continuation $D \rightsquigarrow E$, the recruitment profile $\phi_{D \rightarrow E}(x)$ records where and how strongly substrate degrees of freedom are recruited to carry that continuation. The associated recruitment cost is denoted*

$$E_{\text{rec}}[\phi; \Gamma].$$

Fields are represented equilibrium recruitment profiles. Radiation is cost discharge from failed adiabatic recruitment. Geometry is the special case in which the recruited substrate carries comparison, localization, and transport continuations.

Source of the recruitment formalism. The recruitment profile $\phi_{D \rightarrow E}$, the recruitment cost E_{rec} , and their dynamics are formalized in the Recruitment-Radius papers (Papers 24 and 25): the recruitment cost functional, the master equation for substrate recruitment, continuous anchor profiles, and the substrate-kernel structure. The present paper specializes that formalism to the geometric case — comparison, localization, and transport continuations. Where the derivations below call for a nonlinear recruitment functional, a substrate kernel, or its Green’s function, the source of record is Papers 24/25.

7. Comparison continuations and the three geometric levels

Definition 7.1 (Comparison continuation). *A comparison continuation from x to y in context Γ is a substrate operation $K : x \rightsquigarrow y$ that carries a probe’s held continuation profile from x to y , preserves its complete contextual identity class along the way, and establishes endpoint correspondence. Its ledger may be decomposed as*

$$\kappa_{\Gamma}(K) = \kappa_{\text{transport}}(K) + \kappa_{\text{identity}}(K) + \kappa_{\text{corr}}(K),$$

with the understanding that the three terms may share physical support.

Definition 7.2 (Directed comparison cost and comparison substrate). *The directed comparison cost is*

$$d_{\Gamma}^{+}(x, y) := \inf\{\kappa_{\Gamma}(K) : K \text{ is an admissible comparison continuation } x \rightsquigarrow y\}.$$

A comparison substrate $\mathcal{M}$ is a regular domain of distinguishable comparison sites equipped with admissible nearby comparison continuations. It is locally comparison-complete at scale ℓ when every represented pair below that scale is connected by at least one admissible comparison continuation.

Definition 7.3 (Direct local comparison germ). *Whenever the refinement limit exists, define*

$$c_x^{\text{dir}}(u) := \lim_{\epsilon \downarrow 0} \frac{d_{\Gamma}^{+}(x, x + \epsilon u)}{\epsilon}.$$

This is the direct substrate-side cost germ. It may be asymmetric, nonsmooth, or degenerate before further gates close.

The response construction below introduces three distinct geometries.

Level	Carrier	Role
Process geometry	Represented held-process algebra $\mathfrak{T}_H$ acting on V_H , with operational distance d_{proc}	Measures distinguishability or displacement between represented processes. In the Paper 2 channel model, $\ \Delta\ _{\text{proc}} = \frac{1}{2}\ \Delta\ _{\diamond}$.
Response geometry	Response-null quotient $\mathcal{Q}_{\Pi, \mathcal{F}}$	Measures intrinsic length of changing a represented response class while redundant coordinates have been removed.

Substrate geometry	comparison	Comparison substrate $\mathcal{M}$	Measures comparison, localization, and transport directions at physical sites after an anchor identifies them with response directions.
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Audit Note 7.4 (Type ledger). The following objects must not be silently identified:

object	domain	homogeneity/role
$F_{\text{resp}}(q, v)$	$T\mathcal{Q}$	positive one-homogeneous gauge
$d_{\text{resp}}(q_0, q_1)$	$\mathcal{Q} \times \mathcal{Q}$	intrinsic path distance
I_{Π}^{path}	physical protocol paths	calibrated path ledger
I_{Π}^{res}	endpoint response state	local two-homogeneous residual energy
M_{sep}	$\mathcal{P}$	extrinsic distance to a free-process set
$Q_x^{\text{cmp}}(u)$	$T_x\mathcal{M}$	signed causal comparison function.

A valid theorem may compare these objects, but their equality is never a consequence of notation.

8. Response-null quotient and intrinsic process geometry

Let R_{Π} be a finite-dimensional response-coordinate manifold, fix a free represented held process $\mathcal{F} \in \mathfrak{T}_H$, and let

$$\Phi_{\mathcal{F}} : R_{\Pi} \longrightarrow \mathfrak{T}_H, \quad \Phi_{\mathcal{F}}(0) = \mathcal{F},$$

be a differentiable response model. Coordinates can represent memory occupation, a residual correlation register, a joint repair operation, nonfactorizing control amplitude, or any other finite response variable. Coordinates are not physical merely because they are written down.

Definition 8.1 (Response fiber and regular quotient chart). *For fixed $\mathcal{F}$, define*

$$r \sim_{\mathcal{F}} r' \iff \Phi_{\mathcal{F}}(r) = \Phi_{\mathcal{F}}(r').$$

A regular local response quotient near r_0 is a triple

$$U \xrightarrow{\pi} \mathcal{Q}_{\Pi, \mathcal{F}} \xrightarrow{\bar{\Phi}_{\mathcal{F}}} \mathcal{P}$$

such that π is a surjective submersion, its fibers are exactly the local response fibers, $\Phi_{\mathcal{F}}|_U = \bar{\Phi}_{\mathcal{F}} \circ \pi$, and $\bar{\Phi}_{\mathcal{F}}$ is an immersion.

Proposition 8.2 (Local quotient from constant rank). *Suppose $\Phi_{\mathcal{F}}$ takes values in a finite-dimensional smooth process chart, is C^1 , and has constant rank k on a neighborhood of r_0 . Then after shrinking the neighborhood it admits a regular local response quotient of dimension k . If the rank changes, no smooth global quotient is implied; the natural response geometry may instead be stratified or singular.*

Proof. The constant-rank theorem gives local coordinates (u, w) on the response domain and y on process space in which $\Phi_{\mathcal{F}}(u, w) = (u, 0)$. Projection $(u, w) \mapsto u$ supplies the local quotient, and $u \mapsto (u, 0)$ is the induced immersion. At a rank-changing point this normal form cannot be chosen with fixed quotient dimension, so smooth-manifold conclusions require an additional stratification analysis. $\square$

Definition 8.3 (Pullback process gauge). *Let the represented process displacement space carry a norm $\|\cdot\|_{\text{proc}}$. On a regular quotient define*

$$F_{\text{resp}}(q, v) := \left\| D\bar{\Phi}_{\mathcal{F}, q}(v) \right\|_{\text{proc}} \quad (q \in \mathcal{Q}_{\Pi, \mathcal{F}}, v \in T_q \mathcal{Q}_{\Pi, \mathcal{F}}).$$

For the finite channel model of Paper 2,

$$\|\Delta\|_{\text{proc}} := \frac{1}{2} \|\Delta\|_{\diamond}.$$

Theorem 8.4 (Intrinsic pullback-gauge theorem). *On every regular quotient chart, $F_{\text{resp}}(q, \cdot)$ is a norm on the tangent space. It is invariant under smooth reparametrization of quotient coordinates. If the ambient process norm is not smooth or strictly convex, the result is a possibly nonsmooth Finsler-type structure rather than a classical strongly convex Finsler metric.*

Proof. Positive homogeneity and the triangle inequality follow from the process norm and linearity of $D\bar{\Phi}_q$. If $F_{\text{resp}}(q, v) = 0$, then $D\bar{\Phi}_q v = 0$; immersion of $\bar{\Phi}$ gives $v = 0$. Under a quotient-coordinate change ψ , the chain rule sends v to $D\psi(v)$ and leaves the process tangent vector $D\bar{\Phi}(v)$ unchanged, proving invariance. $\square$

Definition 8.5 (Intrinsic response length and distance). *For an absolutely continuous quotient path $\gamma : [0, 1] \rightarrow \mathcal{Q}_{\Pi, \mathcal{F}}$, define*

$$L_{\text{resp}}[\gamma] := \int_0^1 F_{\text{resp}}(\gamma(t), \dot{\gamma}(t)) dt,$$

and

$$d_{\text{resp}}(q_0, q_1) := \inf_{\gamma: q_0 \rightarrow q_1} L_{\text{resp}}[\gamma],$$

with value $+\infty$ if no admissible path connects the endpoints.

Theorem 8.6 (Endpoint process displacement is bounded by response length). *For every absolutely continuous response path,*

$$d_{\text{proc}}(\bar{\Phi}(q_0), \bar{\Phi}(q_1)) \leq L_{\text{resp}}[\gamma],$$

and therefore

$$d_{\text{proc}}(\bar{\Phi}(q_0), \bar{\Phi}(q_1)) \leq d_{\text{resp}}(q_0, q_1).$$

Here $d_{\text{proc}}(\mathcal{N}_0, \mathcal{N}_1) = \|\mathcal{N}_1 - \mathcal{N}_0\|_{\text{proc}}$ in an affine process chart.

Proof. The image path $\bar{\Phi} \circ \gamma$ is absolutely continuous and has speed $\|D\bar{\Phi}_{\gamma(t)} \dot{\gamma}(t)\|_{\text{proc}} = F_{\text{resp}}(\gamma(t), \dot{\gamma}(t))$. The normed-space triangle inequality bounds endpoint displacement by the integrated speed. Taking the path infimum gives the second inequality. $\square$

Theorem 8.7 (Finite linear chart and Paper 2 recovery). *Suppose*

$$\Phi(\mathcal{F}, r) = \mathcal{F} + T_{\Pi} r$$

on a finite-dimensional affine process chart. Put

$$\bar{R}_{\Pi} := R_{\Pi} / \ker T_{\Pi}, \quad \bar{T}_{\Pi} : \bar{R}_{\Pi} \rightarrow \text{im } T_{\Pi}.$$

Then the quotient response geometry is flat and translation invariant:

$$F_{\text{resp}}([r], [v]) = \|\bar{T}_{\Pi}[v]\|_{\text{proc}},$$

and straight quotient segments minimize length, so

$$d_{\text{resp}}([r_0], [r_1]) = \left\| \bar{T}_{\Pi}([r_1 - r_0]) \right\|_{\text{proc}}.$$

If $\|\cdot\|_{\bar{c}}$ is a calibrated quotient ledger norm, define

$$L_{\Pi}^{\text{opt}} := \max_{\|x\|_{\bar{c}}=1} \|\bar{T}_{\Pi}x\|_{\text{proc}}, \quad \mu_{\Pi} := \min_{\|x\|_{\bar{c}}=1} \|\bar{T}_{\Pi}x\|_{\text{proc}}.$$

For a nontrivial quotient,

$$0 < \mu_{\Pi} \leq L_{\Pi}^{\text{opt}} < \infty,$$

and

$$\frac{1}{L_{\Pi}^{\text{opt}}} d_{\text{resp}}([r_0], [r_1]) \leq \| [r_1 - r_0] \|_{\bar{c}} \leq \frac{1}{\mu_{\Pi}} d_{\text{resp}}([r_0], [r_1]).$$

Proof. The quotient removes exactly the kernel of T_{Π} , so $\bar{T}_{\Pi}$ is injective and the displayed tangent gauge follows from Theorem 8.4. The image of a straight quotient segment is a straight process segment, whose normed-space length equals its endpoint displacement. Theorem 8.6 proves minimality. Compactness of the quotient unit sphere gives the extremal gains, and injectivity makes the minimum positive. Homogeneity gives the final comparison. This is the intrinsic-geometric form of the finite linear response theorem in Paper 2. $\square$

Definition 8.8 (Extrinsic free-process monotone). *For a closed free-process set $\mathfrak{F}_{\text{sep}} \subset \mathcal{P}$, define*

$$M_{\text{sep}}(\mathcal{N}) := \inf_{\mathcal{F} \in \mathfrak{F}_{\text{sep}}} d_{\text{proc}}(\mathcal{N}, \mathcal{F}).$$

Proposition 8.9 (Extrinsic monotone versus intrinsic response distance). *If $\bar{\Phi}(q_0) \in \mathfrak{F}_{\text{sep}}$, then*

$$M_{\text{sep}}(\bar{\Phi}(q_1)) \leq d_{\text{resp}}(q_0, q_1).$$

A reverse bound is not automatic. It requires, for example, that a nearest free process lie in the response chart together with a uniform lower response gain and sufficient quasiconvexity of the response image.

Proof. The free-set infimum is no larger than the distance to the particular free point $\bar{\Phi}(q_0)$, and Theorem 8.6 bounds that process distance by d_{resp} . If the nearest free process lies outside the response image, or if the image has long detours relative to ambient process distance, no converse follows. $\square$

9. Physical-ledger calibration: distance, energy, and the homogeneity fork

Definition 9.1 (Physical interaction-ledger rate). *A physical path ledger on a response quotient is a nonnegative tangent functional $j_q(v)$ whose path cost is*

$$J[\gamma] = \int_0^1 j_{\gamma(t)}(\dot{\gamma}(t)) dt.$$

The induced directed physical path cost is

$$I_{\Pi}^{\text{path}}(q_0, q_1) := \inf_{\gamma: q_0 \rightarrow q_1} J[\gamma].$$

The word “physical” refers to a separately calibrated resource valuation, not merely to the represented process norm.

Theorem 9.2 (Calibrated ledger realization). *Suppose constants $0 < a \leq b < \infty$ satisfy*

$$a F_{\text{resp}}(q, v) \leq j_q(v) \leq b F_{\text{resp}}(q, v)$$

for every admissible tangent direction. Then

$$\boxed{a d_{\text{resp}}(q_0, q_1) \leq I_{\Pi}^{\text{path}}(q_0, q_1) \leq b d_{\text{resp}}(q_0, q_1)}.$$

If $j = F_{\text{resp}}$ exactly, then $I_{\Pi}^{\text{path}} = d_{\text{resp}}$. Equality is therefore a calibration theorem, not a definition of interaction cost.

Proof. Integrate the pointwise bounds along every admissible path and take the infimum. Exact calibration makes the two path functionals identical. $\square$

Boundary 9.3 (When no path metric exists). Fixed setup burdens, packet thresholds, indivisible tokens, saturation, and history-dependent hysteresis can make physical cost fail local one-homogeneity or refinement additivity. In those branches I_{Π}^{path} need not be generated by any tangent length. The response distance remains a valid model geometry, but its identification with the physical ledger fails.

Theorem 9.4 (Local residual energy is squared distance). *Let q_0 be a reference response class and let $\mathcal{R}$ be a C^2 nonnegative residual penalty with*

$$\mathcal{R}(q_0) = 0, \quad D\mathcal{R}(q_0) = 0.$$

Assume the Hessian

$$g_{q_0}^{\text{E}} := D^2\mathcal{R}(q_0)$$

is positive definite and extends to a smooth local Riemannian metric g^{E} . Then

$$\boxed{\mathcal{R}(q) = \frac{1}{2} d_{g^{\text{E}}}(q_0, q)^2 + o(d_{g^{\text{E}}}(q_0, q)^2)}$$

as $q \rightarrow q_0$.

Proof. Choose normal coordinates for g^{E} at q_0 . Taylor expansion gives $\mathcal{R}(q) = \frac{1}{2} g_{q_0}^{\text{E}}(\Delta q, \Delta q) + o(\|\Delta q\|^2)$. In the same coordinates, squared geodesic distance is $g_{q_0}^{\text{E}}(\Delta q, \Delta q) + o(\|\Delta q\|^2)$. Combining the expansions proves the claim. $\square$

Audit Note 9.5 (Interaction cost is not one object). Paper 2 contains both branches. Its quotient coordinate/path-speed bounds are one-homogeneous and naturally define length. Its smooth residual realization gives a two-homogeneous Hessian energy. The correct local dictionary is therefore either

$$I_{\Pi}^{\text{path}} = d_{\text{resp}}$$

under exact path-ledger calibration, or

$$I_{\Pi}^{\text{res}} = \frac{1}{2} d_{g^{\text{E}}}^2 + o(d^2)$$

under smooth residual realization. Writing $I_{\Pi} = d_g$ without specifying the branch loses this distinction.

10. Anchoring response geometry to the comparison substrate

Response geometry lives on implementation classes. To obtain substrate geometry one needs a physical map from comparison directions to response directions.

Definition 10.1 (Universal infinitesimal comparison family). *Fix a regular region $U \subseteq \mathcal{M}$. For each $x \in U$, let $\mathcal{A}_x$ denote the full declared family of admitted infinitesimal comparison continuations based at x . Every $a \in \mathcal{A}_x$ has a linearized response map*

$$\ell_{a,x} : T_x \mathcal{M} \longrightarrow Y_{a,x},$$

where $Y_{a,x}$ is the corresponding infinitesimal outcome or process-response space. Define the universal comparison evaluation map

$$E_x : T_x \mathcal{M} \longrightarrow \prod_{a \in \mathcal{A}_x} Y_{a,x}, \quad E_x u := (\ell_{a,x} u)_{a \in \mathcal{A}_x}.$$

No topology on the full product is needed for the kernel statements below. On a finite-rank regular branch one may replace the full family by any finite separating subfamily.

Definition 10.2 (Contextual null distribution). *Two tangent directions are contextually equivalent exactly when every admitted infinitesimal comparison gives the same response:*

$$u \approx_x v \iff \ell_{a,x}(u - v) = 0 \text{ for every } a \in \mathcal{A}_x.$$

Therefore the contextual null space is not an independent declaration but the kernel of the universal comparison evaluation:

$$N_x^{\text{ctx}} := \ker E_x = \bigcap_{a \in \mathcal{A}_x} \ker \ell_{a,x}.$$

The contextual tangent quotient is

$$T_x^{\text{ctx}} \mathcal{M} := T_x \mathcal{M} / N_x^{\text{ctx}}.$$

When $x \mapsto N_x^{\text{ctx}}$ has locally constant rank, these fibers form a smooth quotient bundle.

Definition 10.3 (Comparison and physical-response germs). *Let $\mathcal{C}_x$ be the set of admissible C^1 curve germs $[\gamma]_x$ through x . Each admitted comparison $a \in \mathcal{A}_x$ assigns an outcome germ*

$$\mathcal{L}_{a,x} : \mathcal{C}_x \longrightarrow \mathcal{Y}_{a,x}.$$

The complete comparison-germ evaluation is

$$\mathcal{E}_x([\gamma]_x) := (\mathcal{L}_{a,x}([\gamma]_x))_{a \in \mathcal{A}_x}.$$

Let

$$\mathcal{B}_x^{\text{phys}} : \mathcal{C}_x \longrightarrow \mathcal{P}_x^{\text{red}}$$

be the physical process-response germ valued in the reduced response carrier, obtained after quotienting gauge, auxiliary, and otherwise record-invisible representation variables. Thus $\mathcal{P}_x^{\text{red}}$ contains only distinctions that are carried by some physical record sector.

Theorem 10.4 (Closed-world germ-completeness theorem). *Assume the closed-world closure imported from Papers 0 and 1:*

- (i) every finite enforceable distinction belongs to the closed interface ledger and is therefore readable in the complete reduced process response;

(ii) no stable reduced process direction is absent from every record sector.

Then the complete comparison-germ evaluation and the reduced physical-response germ have the same fibers:

$$\boxed{\mathcal{B}_x^{\text{phys}}([\gamma]_x) = \mathcal{B}_x^{\text{phys}}([\eta]_x) \iff \mathcal{E}_x([\gamma]_x) = \mathcal{E}_x([\eta]_x).}$$

Consequently both maps factor through one and the same contextual quotient

$$\mathcal{Q}_x := \mathcal{C}_x / \sim_{\text{ctx}}, \quad [\gamma]_x \sim_{\text{ctx}} [\eta]_x \iff \mathcal{E}_x([\gamma]_x) = \mathcal{E}_x([\eta]_x),$$

and the induced reduced physical carrier is canonically isomorphic to $\mathcal{Q}_x$ by the unique class-preserving bijection.

Proof. Suppose first that the reduced physical-response germs agree. Any admitted comparison distinguishes a finite cost-bearing partition and therefore belongs to the closed interface ledger. By closed-world closure its outcome is a readout of the complete reduced process response. Equal reduced response germs therefore give equal outcome germs for every a , so the comparison evaluations agree.

Conversely suppose the comparison evaluations agree but the reduced physical response germs differ. Their difference is then a stable distinction in the reduced process carrier. Because this carrier has already been quotiented by record-invisible redundancy, the distinction must be carried by at least one record sector. The corresponding finite continuation is an admitted comparison in the complete family $\mathcal{A}_x$, and it would distinguish the two comparison evaluations, contradiction. Thus the fibers coincide.

Maps with the same fibers induce bijections from their common quotient onto their images. Since $\mathcal{P}_x^{\text{red}}$ is defined as the complete reduced image, the class-preserving bijection is onto and unique. $\square$

Audit Note 10.5 (What closed-world completeness contributes). The theorem does not infer derivative faithfulness directly from ledger closure. It first proves the stronger and correctly typed statement at the level at which Paper 0 defines physical identity: complete families of finite admissible continuations. This avoids the false inference that global separation automatically implies injective differentiation. The latter fails, for example, for $t \mapsto t^2$ at $t = 0$.

Definition 10.6 (Finite-stage record test). *Let $W \subseteq P^{\text{red}}$ be a comparison sector, with no topology assumed. A finite-stage record test on W is a map*

$$r_k : W \longrightarrow S_k,$$

where S_k is a finite outcome set and r_k is produced by a finite admissible continuation and deposited in a physical record sector. The fibres of all such tests form a Boolean algebra of finite operational distinctions. They are not, merely by being executable, declared open sets.

Definition 10.7 (Precision-refinement channel and asymptotic quotient). *A precision-refinement channel is a compatible tower*

$$\mathbf{r} = (r_k, \pi_{k+1,k})_{k \geq 0}, \quad r_k : W \rightarrow S_k, \quad r_k = \pi_{k+1,k} \circ r_{k+1},$$

with every S_k finite. A real realization consists of encodings $\iota_k : S_k \rightarrow \mathbb{R}$ for which $\iota_k r_k$ is locally uniformly Cauchy with certified error $\epsilon_k \downarrow 0$. Its limit is

$$\hat{r} = \lim_{k \rightarrow \infty} \iota_k r_k.$$

Two complete tower histories are asymptotically record-equivalent when their encoded separation tends to zero. The realization is quotient-faithful when equality of real limits is exactly asymptotic record-equivalence. Thus finite execution is retained at every precision while histories that differ only by vanishing boundary conventions are identified in the reduced channel.

Proposition 10.8 (Unreduced tower-faithfulness no-go). *Let $X_{\mathbf{r}}$ be the inverse limit of the finite discrete stage spaces. If a real realization is continuous and injective on the unreduced tower space $X_{\mathbf{r}}$, then its image is compact and zero-dimensional. A finite product of such tower-faithful channel images is again zero-dimensional and cannot contain a nontrivial connected smooth continuation orbit.*

Proof. A countable inverse limit of finite discrete spaces is compact, metrizable, and zero-dimensional. A continuous injection from a compact space into Hausdorff $\mathbb{R}$ is a homeomorphism onto its image, so the image remains zero-dimensional. Finite products preserve zero-dimensionality. Every connected subset is therefore a singleton. $\square$

Corollary 10.9 (Positive-dimensional smooth continuum requires more than faithful tower relabelling). *A continuous injective real relabelling of an unreduced profinite tower may exist, but its image remains zero-dimensional. Therefore no finite product of such faithful relabellings can by itself contain a nontrivial connected positive-dimensional smooth continuation orbit. The physical construction must either pass to an asymptotic quotient that changes the relevant topology or add independently certified non-profinite smooth structure. Real-valued faithful embeddings of Cantor-type tower spaces are not excluded.*

Definition 10.10 (Record-limit topology). *Let $\mathcal{R}_{\text{ch}}(W)$ be the family of ledger-certified real channel limits on W . The record-limit topology τ_{rec} is the initial topology for the maps*

$$\hat{r} : W \longrightarrow \mathbb{R}, \quad \hat{r} \in \mathcal{R}_{\text{ch}}(W),$$

where $\mathbb{R}$ carries its ordinary topology. Equivalently, a subbasis is formed by

$$\hat{r}^{-1}((a, b)), \quad a, b \in \mathbb{Q}, \quad a < b.$$

Finite-stage fibres certify finite-resolution distinctions and approximate these neighbourhoods; they are not automatically clopen in τ_{rec} .

Boundary 10.11 (Why the finite-stage topology is rejected). If every fibre of every finite-stage test were declared open, the resulting Hausdorff refinement topology would be zero-dimensional and totally disconnected. It could not contain a nontrivial connected smooth continuation orbit. APF therefore uses the finite stages as executable approximants and constructs physical neighbourhoods from the ordered limits of precision channels. This is the exact place where the continuum topology enters; it is not the unrestricted profinite completion of the finite partitions.

Proposition 10.12 (Finite-stage clopen no-go). *Let τ_{fin} be the topology generated by declaring every finite-stage record fibre open. Then every such fibre is clopen. If the finite stages separate points, (W, τ_{fin}) is Hausdorff and zero-dimensional; hence all connected subsets are singletons. No positive-dimensional connected smooth orbit can be embedded continuously in this topology.*

Proof. Each stage has finitely many fibres, so the complement of any one fibre is the finite union of the others and is open. Thus the fibre basis is clopen and the space is zero-dimensional. A Hausdorff zero-dimensional space is totally disconnected. The continuous image of a connected manifold is connected, so an injective continuous orbit inclusion would have singleton image unless the orbit were zero-dimensional. $\square$

Proposition 10.13 (Closed-world Hausdorff separation). *If the complete real channel family separates reduced physical states, then $(P^{\text{red}}, \tau_{\text{rec}})$ is Hausdorff.*

Proof. Let $p \neq q$. Closed-world reduction supplies a real precision channel $\hat{r}$ with $\hat{r}(p) \neq \hat{r}(q)$. Choose disjoint rational open intervals containing the two values. Their inverse images are disjoint τ_{rec} -neighbourhoods of p and q . $\square$

Assumption 10.14 (Countable channel-specification grammar). *The primitive instruction tokens, finitely encoded exact parameters, stopping rules, bonding-map descriptions, encodings, and error certificates available on the declared interface form a countable specification language. Arbitrary real constants are not admitted as single primitive tokens without an explicit countable code.*

Definition 10.15 (Finitely specified physical channel). *A physical precision channel is finitely specified when one finite word in the grammar of Assumption 10.14 specifies its stage tests, bonding maps, requested-precision stopping rule, encodings, and certified error bound.*

Theorem 10.16 (Countable specified channel family). *Under Assumption 10.14, the finitely specified physical precision channels on a fixed closed interface form a countable family.*

Proof. Finite words over a countable alphabet form a countable set. $\square$

Gate 10.17 (Countable topology generation). *The finitely specified quotient-faithful real limits contain a countable subfamily whose initial topology equals the record-limit topology of the complete ledger-certified channel family.*

Corollary 10.18 (Conditional second countability). *Under Assumption 10.14 and Gate 10.17, the record-limit topology is second countable.*

Proof. Inverse images of rational intervals under the countable topology-generating subfamily form a countable subbasis, and finite intersections form a countable base. $\square$

Lemma 10.19 (Finite stages approximate record-limit neighbourhoods). *Let $\hat{r}$ be a channel limit with certified local error $|\iota_k r_k - \hat{r}| \leq \epsilon_k$. For every p and every open interval $I \ni \hat{r}(p)$, there are a smaller interval J with $\hat{r}(p) \in J \subseteq I$ and a stage k such that the finite-stage cell containing p is mapped by ι_k into the ϵ_k -thickening of J , which lies in I . Hence every record-limit neighbourhood is operationally resolvable to finite certified precision.*

Proof. Choose J with positive distance δ from its closure to $\mathbb{R} \setminus I$. Take k with $\epsilon_k < \delta/2$ and with $\iota_k r_k(p)$ within ϵ_k of $\hat{r}(p)$. Compatibility and the certified error bound place all states in the corresponding stage cell whose encoded value lies in the chosen thickening inside I . No claim that the entire stage cell is open is required. $\square$

Definition 10.20 (Executable channel algebra). *For $U \in \tau_{\text{rec}}$, let $\mathcal{A}_{\text{ch}}(U)$ be the unital real algebra generated by restrictions of real precision-channel limits and by ledger-certified locally uniform limits of executable scalar postprocessings. Every generator is τ_{rec} -continuous by construction or by uniform convergence. Let $\mathcal{A}_0(U)$ denote its closure under such locally uniform limits.*

Audit Note 10.21 (Topology ledger). The finite stages provide the finite executable evidence. The real channel limits provide the ordered topology. Closed-world completeness provides point separation. A countable specification grammar provides a countable specified channel family; second countability follows only when that family is topology-generating. None of these statements identifies the raw inverse limit of finite partitions with a continuum, and none assumes a manifold.

Definition 10.22 (Executable postprocessing and smooth saturation). *Let $f = (f_1, \dots, f_m)$ be a jointly realizable finite channel map at the requested precision. A scalar postprocessing is executable when it is given by a finite protocol at each requested precision. It is observationally conservative when it factors through f , and therefore cannot distinguish two states that f does not already distinguish.*

The smooth saturation of a family of real readouts is its closure under all maps $\psi \in C^\infty(\mathbb{R}^m)$. Smooth saturation is a mathematical closure of the represented observable algebra; it does not assert that every smooth map is itself a primitive finite program.

Theorem 10.23 (Lossless postprocessing factor theorem). *Every executable deterministic postprocessing of a closed joint record is observationally conservative. Conversely, adjoining any scalar function already constant on the fibres of the closed joint record changes neither those fibres nor the closed-world contextual quotient. No executability claim is implied in this converse.*

Proof. If $h = \varphi \circ f$, then $f(p) = f(q)$ implies $h(p) = h(q)$, so h creates no new distinction class. Appending an executable protocol to the record continuation changes implementation cost but not the partition of physical states induced by the complete record. Conversely, any scalar function constant on every fibre of f factors set-theoretically through the image of f ; adjoining it leaves those fibres unchanged. Hence the contextual quotient is invariant under such lossless relabelling. $\square$

Definition 10.24 (Refinement-complete APF differential sheaf). *For $U \in \tau_{\text{rec}}$, let $\mathcal{R}(U)$ contain the refinement-stable real readouts obtained from executable records and their ledger-certified local limits. Define $\mathcal{C}_{\text{APF}}(U)$ to be the localization and smooth saturation of $\mathcal{R}(U)$: it is the smallest family of real functions on U that*

- (i) contains $\mathcal{R}(U)$;
- (ii) is closed under restriction and local gluing;
- (iii) satisfies
$$f_1, \dots, f_m \in \mathcal{C}_{\text{APF}}(U), \quad \psi \in C^\infty(\mathbb{R}^m) \implies \psi(f_1, \dots, f_m) \in \mathcal{C}_{\text{APF}}(U).$$

Theorem 10.25 (Differential-space closure theorem). *The triple*

$$(P^{\text{red}}, \tau_{\text{rec}}, \mathcal{C}_{\text{APF}})$$

is a Sikorski differential space. Its topology is the initial topology of the physical readouts, and its differential structure is closed under smooth superposition and localization.

Proof. The record-limit topology is, by Definition 10.10, the initial topology of the generating physical channel limits. Smooth superpositions of continuous generators remain continuous, and localization adds only functions that locally agree with such continuous functions. Definition 10.24 is therefore closed under smooth superposition and localization and induces the same topology. These are the Sikorski axioms. Proposition 10.13 and Proposition 10.18 supply Hausdorffness and, on countably generated sectors, second countability; neither property is needed to define the differential structure itself. The lossless-postprocessing factor theorem shows that the saturation changes no closed-world record fibre. $\square$

Audit Note 10.26 (What is derived and what is saturated). The executable channel algebra, its record-limit topology, and invariance of record fibres under deterministic postprocessing are derived from APF closure. Closure under the full C^∞ functional calculus is the canonical smooth saturation

of that algebra. It is not disguised computability and it does not assert that finite records alone force a manifold. Its role is to define the minimal differential structure compatible with the already represented readouts; manifold structure appears only on the regular continuation orbits below.

Definition 10.27 (Derivation tangent and algebraic cotangent). *Let $\mathcal{C}_p$ be the stalk of $\mathcal{C}_{\text{APF}}$ at p , and let $\mathfrak{m}_p = \{[f]_p : f(p) = 0\}$ be its maximal ideal. The primary tangent is*

$$T_p^{\text{der}} P := \text{Der}_p(\mathcal{C}_p, \mathbb{R}),$$

the real derivations satisfying Leibniz' rule. The algebraic first-order cotangent candidate is

$$T_p^{*,\text{alg}} P := \mathfrak{m}_p / \mathfrak{m}_p^2.$$

Every derivation defines a linear functional on $\mathfrak{m}_p / \mathfrak{m}_p^2$; Proposition 10.28 shows that this pointwise pairing is canonical and perfect. Singular behaviour appears through rank variation and failure of finite local presentation.

Proposition 10.28 (Canonical tangent–cotangent duality). *Evaluation on first-order classes gives a canonical vector-space isomorphism*

$$T_p^{\text{der}} P \cong (\mathfrak{m}_p / \mathfrak{m}_p^2)^*, \quad X \mapsto ([f] \mapsto Xf).$$

Singularity is therefore detected by failure of finite local presentation, rank jumps, or variation of this common dimension—not by a spurious gap between the two objects.

Proof. Leibniz' rule makes every derivation vanish on $\mathfrak{m}_p^2$, so the map is well defined and injective. Conversely, for $\lambda \in (\mathfrak{m}_p / \mathfrak{m}_p^2)^*$ define

$$X_\lambda(f) := \lambda([f - f(p)]).$$

Modulo $\mathfrak{m}_p^2$ one has $fg - f(p)g(p) = f(p)(g - g(p)) + g(p)(f - f(p))$, hence X_λ satisfies Leibniz' rule at p . The two constructions are inverse. $\square$

Definition 10.29 (Finite differential presentation). *A record-open neighbourhood U has a finite differential presentation when there exist real channel limits $f_1, \dots, f_N \in \mathcal{C}_{\text{APF}}(U)$ such that*

(i) the evaluation map

$$F_U = (f_1, \dots, f_N) : U \rightarrow \mathbb{R}^N$$

is injective and is a homeomorphism onto its image;

(ii) every $g \in \mathcal{C}_{\text{APF}}(U)$ is locally of the form $g = \psi \circ F_U$ for some $\psi \in C^\infty(\mathbb{R}^N)$.

The locus of points admitting such a neighbourhood is denoted P^{fdiff} .

Definition 10.30 (Ledger-independent standing channels). *A jointly realizable family of precision channels $\mathbf{r}^1, \dots, \mathbf{r}^m$ is ledger-independent on U when no complete tower $\mathbf{r}^i$ is determined by the joint complete tower of the others. Equivalently, for every i there exist reduced states $p_i, q_i \in U$ that agree in all channels $j \neq i$ but differ in channel i . A ledger-independent channel therefore carries a distinction not recoverable from the remaining standing commitments.*

Theorem 10.31 (Additive channel-capacity bound). *Assume the APF positive standing-cost floor $\varepsilon_* > 0$, additive ledger accounting for jointly realized ledger-independent commitments, and finite local capacity $C(U) < \infty$. Then every ledger-independent jointly realizable family of precision channels on U has cardinality*

$$m \leq \left\lfloor \frac{C(U)}{\varepsilon_*} \right\rfloor.$$

The bound concerns the number of independent standing channels, not the number of finite-stage outcomes, the number of states, or the cardinality of any refinement limit.

Proof. Ledger independence assigns to each channel a distinction unavailable from the others. By the positive floor, maintaining each such standing distinction costs at least ε_* . Additivity for simultaneously maintained independent commitments therefore gives total standing charge at least $m\varepsilon_*$. Admissibility requires $m\varepsilon_* \leq C(U)$. $\square$

Definition 10.32 (Complete channel basis). *A complete channel basis on U is a jointly realizable channel family that separates reduced states and is inclusion-minimal with that property. Equivalently, deleting any one channel destroys separation.*

Lemma 10.33 (Minimal completeness implies ledger independence). *Every complete channel basis is ledger-independent.*

Proof. If channel i were determined by all the others, then equality in the remaining channels would force equality in channel i . Deleting channel i would therefore preserve the joint fibres and hence preserve separation, contradicting inclusion-minimality. $\square$

Theorem 10.34 (Finite-channel completeness). *Let U admit a complete channel basis and satisfy the additive channel-capacity theorem. Then every complete channel basis is finite:*

$$\mathbf{F}_U = (\mathbf{r}^1, \dots, \mathbf{r}^N), \quad N \leq \left\lfloor \frac{C(U)}{\varepsilon_*} \right\rfloor.$$

Its joint real limit map

$$\hat{F}_U = (\hat{r}^1, \dots, \hat{r}^N) : U \rightarrow \mathbb{R}^N$$

is injective whenever the chosen channels possess faithful real realizations.

Proof. By Lemma 10.33, a complete channel basis is ledger-independent. The additive capacity bound therefore makes it finite. Completeness says that equality of every complete channel tower implies equality in the reduced carrier. Faithful real realization preserves each tower's complete fibres; hence equality of all real limits implies equality of all channel towers and therefore equality of states. Thus $\hat{F}_U$ is injective. $\square$

Gate 10.35 (Finite closed-interface generation). *Finite capacity bounds every jointly realizable ledger-independent channel family, but it does not by itself prove that one finite family separates the entire reduced sector. The missing statement is the finite-generation theorem*

closed-world completeness + finite capacity $\implies$ finite complete physical channel basis.

It requires either a channel-dependence exchange theorem, a compact finite-subcover theorem for physical distinctions, or an equivalent APF generation result. Without one of these, a countable separating family can require infinitely many channels even though no large independent family is jointly realizable.

Proposition 10.36 (Finite-channel sufficiency versus finite differential sufficiency). *Theorem 10.34 gives a finite separating channel map $\hat{F}_U$. It gives a finite differential presentation exactly when*

- (i) $\hat{F}_U$ is a homeomorphism from U onto its image in the record topology; and
- (ii) every $g \in \mathcal{C}_{\text{APF}}(U)$ factors locally as $g = \psi \circ \hat{F}_U$ with $\psi \in C^\infty(\mathbb{R}^N)$.

Proof. Separation gives injectivity. The topology and smooth-factorization clauses are precisely the two additional conditions in Definition 10.29. No finite-image inference is used. $\square$

Theorem 10.37 (Subcartesian realization criterion). *Every point of P^{fdiff} has a neighbourhood isomorphic, as a differential space, to a differential subspace of a subset of $\mathbb{R}^N$. Hence P^{fdiff} is locally subcartesian. Conversely, any locally subcartesian presentation by physical readouts is a finite differential presentation in the sense above.*

Proof. Condition (i) identifies U homeomorphically with $F_U(U)$. Condition (ii) says that the differential structure on U is exactly the pullback of the restriction of $C^\infty(\mathbb{R}^N)$ to that image. This is the defining local model of a subcartesian differential space. The converse is immediate from a finite ambient-coordinate presentation. $\square$

Gate 10.38 (Finite smooth generation). *Even after a finite state-separating channel family is obtained, finite smooth generation is a distinct theorem. It requires that the continuation-regular readout sheaf be locally the smooth functional calculus of that finite map:*

$$\mathcal{C}_{\text{APF}}|_U = \text{Loc } C^\infty(f_1, \dots, f_N).$$

State separation alone does not imply this equality. Raw threshold records may be discontinuous, and continuous limit readouts need not be smooth functions of a separating map. The subcartesian orbit theorem is invoked only after this finite smooth-generation statement closes.

Audit Note 10.39 (Last-open-item ledger). The topology layer now has no hidden certificate: quotient-faithful channel limits, Hausdorff separation, and second countability are derived. Two substantive APF theorems remain before smooth finite-dimensional geometry is unconditional:

- (i) finite closed-interface generation of a state-separating physical channel basis;
- (ii) finite smooth generation of the continuation-regular readout sheaf by that basis.

These are not notation problems. They are the exact remaining continuum and regularity content.

Remark 10.40 (Imported singular differential geometry). For subcartesian differential spaces, derivations satisfy the smooth chain rule. The orbits of a family complete in the orbit-theorem sense carry unique smooth manifold structures for which the orbit inclusions are smooth immersions. The paper imports these established results rather than reproving them. See Sikorski's differential-space formalism and the subcartesian orbit results of Śniatycki and collaborators.¹

Definition 10.41 (Continuation distribution and geometric orbits). *Let $\mathcal{X}_{\text{adm}}$ be a family of vector fields on P^{fdiff} , where a vector field is a derivation that generates a local one-parameter group of local diffeomorphisms. The family is complete in the orbit-theorem sense when for every $X, Y \in \mathcal{X}_{\text{adm}}$ and every time t for which the local flow φ_t^X is defined, the pushforward $(\varphi_t^X)_*Y$ agrees locally with a member of $\mathcal{X}_{\text{adm}}$. Equivalently, the family is closed locally under the flow transports used to form its pseudogroup. Define*

$$\Delta_p := \text{span}\{X(p) : X \in \mathcal{X}_{\text{adm}}\} \subseteq T_p^{\text{der}}P.$$

The orbit through p under finite compositions of these local flows is denoted $\mathcal{O}_p$.

¹J. Śniatycki, *Orbits of families of vector fields on subcartesian spaces*, Ann. Inst. Fourier 53 (2003), 2257–2296, DOI: 10.5802/aif.2006; J. Śniatycki, *Differential Geometry of Singular Spaces and Reduction of Symmetry*, Cambridge University Press (2013), DOI: 10.1017/CBO9781139136990; R. Cushman and J. Śniatycki, *On subcartesian spaces* *Leibniz' rule implies the chain rule*, Can. Math. Bull. 63 (2020), DOI: 10.4153/S0008439519000513.

Theorem 10.42 (Smooth geometric-orbit theorem). *Every orbit $\mathcal{O}_p$ of an admissible continuation family complete in the orbit-theorem sense on P^{fdiff} carries a unique smooth manifold structure for which its inclusion into the differential space is a smooth immersion. Its tangent space is the continuation distribution:*

$$T_q \mathcal{O}_p = \Delta_q \quad (q \in \mathcal{O}_p).$$

Thus smooth manifolds arise as the dynamically connected regular pieces of the APF differential space, not as prior substrate assumptions.

Proof. By Theorem 10.37, P^{fdiff} is locally a subcartesian differential space. The source-level completeness clause in Definition 10.41 is exactly the closure hypothesis required by the orbit theorem for complete families of vector fields on subcartesian spaces; hence it applies to $\mathcal{X}_{\text{adm}}$ and makes each orbit a smooth manifold with the stated tangent distribution. $\square$

Definition 10.43 (Maximal smooth geometric branch and singular locus). *A smooth geometric branch is a maximal connected continuation orbit on which $\dim \Delta$ is locally constant and positive. The union of all such orbits is $P_{\text{reg}}^{\text{geom}}$. Its complement*

$$P^{\text{sing}} := P^{\text{red}} \setminus P_{\text{reg}}^{\text{geom}}$$

is the intrinsic singular locus. It includes zero-rank, rank-jump, saturation, branching, and non-flowable points.

Theorem 10.44 (Existence of the regular geometric regime). *Let $p \in P^{\text{fdiff}}$. If the admissible continuation family is complete in the orbit-theorem sense near p , continuation-active at p , and has locally constant positive orbit rank, then there is a finite-dimensional smooth geometric branch through p .*

Proof. The subcartesian realization criterion supplies the required local singular smooth model. Continuation activity gives a nonzero flow-generating vector field, and Theorem 10.42 gives an immersed smooth orbit through p . Local constancy of $\dim \Delta$ fixes its dimension, while maximal connected extension produces the branch. $\square$

Theorem 10.45 (Consolidated APF regular-geometry theorem). *Let $U \subseteq P^{\text{red}}$ be a closed-world comparison domain satisfying: a quotient-faithful state-separating real channel family; the countable grammar and topology-generation conditions; a complete finite channel basis; finite smooth generation; and a locally flowable continuation family complete in the orbit-theorem sense. Then:*

- (i) *the channel limits generate a Hausdorff second-countable record topology;*
- (ii) *localization and smooth saturation produce a Sikorski differential space;*
- (iii) *on $U \cap P^{\text{fdiff}}$ the space is locally subcartesian;*
- (iv) *every orbit of the complete admissible-continuation family is an immersed smooth manifold; and*
- (v) *on each constant-rank orbit S satisfying Gate 10.50,*

$$\ker D\beta_x = N_x^{\text{ctx}} = \bigcap_{a \in \mathcal{A}_x} \ker \ell_{a,x}, \quad x \in S,$$

with injective induced map

$$T_x S / N_x^{\text{ctx}} \hookrightarrow T_{\beta(x)} \mathcal{O}_{\beta(x)}.$$

Thus smooth response geometry is the regular orbit locus of the APF record differential space, not an upstream manifold postulate.

Proof. Item (i) is Propositions 10.13 and 10.18. Items (ii)–(iv) are Theorem 10.25, Theorem 10.37, and Theorem 10.42. Item (v) is Theorem 10.51 and Corollary 10.52. $\square$

Audit Note 10.46 (Resolution of the continuum-emergence objection). The raw refinement completion remains profinite and may be totally disconnected. Smooth continuum is not obtained by declaring its points close. It appears on the finite-differential-presentation locus as an orbit of physically admissible local continuation derivations. Thus two independent conditions are visible rather than hidden: smooth local factorization of the record algebra, and continuation-active local flows. Failure of either leaves a singular, higher-order, or zero-dimensional record sector rather than a smuggled manifold.

Audit Note 10.47 (Fortification ledger: what remains conditional). The differential-space architecture is now noncircular, but four obligations remain deliberately named rather than hidden:

- (i) the complete physical channel family must admit faithful realizations and separate reduced states;
- (ii) the sector must admit a countable topology-generating channel subfamily when second countability is claimed;
- (iii) finite-channel Regime R must supply a complete channel basis, while the capacity theorem proves only that such a basis is finite;
- (iv) the finite joint channel map must satisfy the homeomorphism and smooth local-factorization clauses before the subcartesian orbit theorem is invoked.

These are falsifiable regime certificates. None is replaced by the statement that finite partitions alone produce a continuum.

Definition 10.48 (Contextual comparison differentials). *On a smooth geometric branch S , every admitted regularized comparison $L_a \in \mathcal{C}_{\text{APF}}$ has differential*

$$\ell_{a,x} := d(L_a|_S)_x : T_x S \rightarrow Y_{a,x}.$$

Define

$$N_x^{\text{ctx}} := \bigcap_{a \in \mathcal{A}_x} \ker \ell_{a,x}.$$

Let $\beta : S \rightarrow P^{\text{red}}$ be the complete reduced physical-response map and $B_x^{\text{phys}} := D\beta_x$ on the orbit manifold.

Audit Note 10.49 (Same fibres do not imply the same derivative kernel). On $\mathbb{R}$, $L(t) = t$ and $\beta(t) = t^3$ have identical fibres, but $dL_0 \neq 0$ and $D\beta_0 = 0$. Set-level closed-world completeness is therefore insufficient for first-order faithfulness.

Gate 10.50 (Smooth response-factor completeness). *On a neighbourhood $V \subseteq S$ of x , there is a finite APF readout generator*

$$F = (f_1, \dots, f_N) : V \rightarrow \mathbb{R}^N$$

whose components are admitted readouts and generate the local Sikorski structure, together with smooth maps on the relevant images such that

$$F = \Psi \circ \beta, \quad \beta = \Theta \circ F.$$

Theorem 10.51 (Regular-orbit closed-world kernel theorem). *On every smooth geometric branch satisfying closed-world germ completeness and Gate 10.50,*

$$\ker D\beta_x = \ker DF_x = \bigcap_{a \in \mathcal{A}_x} \ker \ell_{a,x} = N_x^{\text{ctx}}.$$

Proof. Differentiating $F = \Psi \circ \beta$ and $\beta = \Theta \circ F$ gives $\ker D\beta_x = \ker DF_x$. Every admitted readout is locally a smooth function of F , so $\ker DF_x$ lies in every admitted readout kernel. Conversely the components of F are admitted readouts, so a vector annihilated by all admitted readout differentials lies in $\ker DF_x$. $\square$

Corollary 10.52 (Faithful quotient response). *The induced map*

$$\tilde{B}_x : T_x S / N_x^{\text{ctx}} \longrightarrow T_{\beta(x)} \mathcal{O}_{\beta(x)}, \quad [u] \longmapsto B_x^{\text{phys}} u,$$

is injective.

Definition 10.53 (Universal operational response carrier). *For $x \in S$, define*

$$T_x^{\text{op}} S := T_x S / N_x^{\text{ctx}}.$$

This quotient is universal among linear first-order response realizations that annihilate every contextually null direction.

Theorem 10.54 (Exact-realization criterion for a concrete response model). *Let $B_x^{\text{raw}} : T_x S \rightarrow W_x$ be a concrete finite response model. The following are equivalent:*

- (i) $\ker B_x^{\text{raw}} = N_x^{\text{ctx}}$;
- (ii) B_x^{raw} factors through an injective map $T_x^{\text{op}} S \hookrightarrow W_x$;
- (iii) *the model is sound on contextual null directions and jointly separating on the quotient.*

Proof. This is the first isomorphism theorem applied to the quotient map $T_x S \rightarrow T_x S / N_x^{\text{ctx}}$. $\square$

Theorem 10.55 (Canonical response-orbit quotient gauge). *Let S be a smooth geometric branch and suppose the process carrier has norm $\|\cdot\|_{\text{proc}}$. Then*

$$c_x^{\text{op}}([u]) := \|\tilde{B}_x[u]\|_{\text{proc}}$$

defines a positive homogeneous gauge on $T_x^{\text{op}} S = T_x S / N_x^{\text{ctx}}$. If the response norm is smooth and strongly convex away from zero, these gauges form a Finsler structure on each locally constant-rank orbit stratum. This gauge is intrinsic to the response orbit; it is not yet a map from physical site displacements.

Proof. Well-definedness follows from quotienting by the kernel; positivity follows from injectivity of $\tilde{B}_x$. Homogeneity and convexity pull back from the process norm. Smooth bundle dependence follows on constant-rank orbit strata. $\square$

Corollary 10.56 (Direct pullback case). *When β itself is the complete response map, the operational gauge is the direct pullback of the process norm by $D\beta$ after quotienting its kernel.*

Theorem 10.57 (Intrinsic response-orbit length theorem). *For every piecewise C^1 curve δ in a constant-rank smooth response orbit,*

$$L_{\text{orb}}[\delta] = \int c_{\delta(t)}^{\text{op}}([\dot{\delta}(t)]) dt = L_{\text{proc}}[\beta \circ \delta].$$

Consequently endpoint process displacement is bounded by intrinsic response-orbit length, and the induced point function is a pseudometric before quotienting zero-length connectivity.

Proof. The chain rule gives $D\beta_{\delta(t)}\dot{\delta}(t) = (\beta \circ \delta)'(t)$. Substitute into the definition of c^{op} and integrate. $\square$

Gate 10.58 (Physical comparison-substrate anchor). *There exist a physical regular comparison domain M_{reg} , a smooth site map $s : M_{\text{reg}} \rightarrow S$, and a locally constant-rank bundle map*

$$A_x : T_x M_{\text{reg}} \rightarrow T_{s(x)}^{\text{op}} S$$

such that, with $q_y : T_y S \rightarrow T_y^{\text{op}} S$ the quotient map:

- (i) $\ker A_x$ is exactly the physical comparison-null subspace;
- (ii) A_x has declared process/ledger provenance and is response-gauge invariant;
- (iii) $A_x = q_{s(x)} \circ Ds_x$;
- (iv) $c_x^{\text{dir}}(u) = c_{s(x)}^{\text{op}}(A_x u)$; and
- (v) the calibration is stable on the regular branch.

Theorem 10.59 (Conditional physical substrate pullback). *If Gate 10.58 holds, define $c_x^{\text{sub}}(u) := c_{s(x)}^{\text{op}}(A_x u)$. Then every piecewise C^1 curve γ in M_{reg} satisfies*

$$L_{\text{sub}}[\gamma] := \int c_{\gamma(t)}^{\text{sub}}(\dot{\gamma}(t)) dt = L_{\text{orb}}[s \circ \gamma].$$

Proof. Gate 10.58(iii) gives $A_{\gamma(t)}\dot{\gamma}(t) = [(s \circ \gamma)'(t)]$. Equality of the integrands, and hence of the lengths, follows. $\square$

Proposition 10.60 (Positive cost metric from quadraticity). *If the process norm satisfies the parallelogram identity on the image of every $\tilde{B}_x$, then polarization defines a positive-definite inner product on $T_x^{\text{op}} S$. These inner products form the positive cost metric g^{cost} on each smooth constant-rank branch.*

Proof. The Jordan–von Neumann theorem turns the quotient norm into an inner product. Injectivity makes it positive definite; smooth dependence follows from the smooth response representation. $\square$

Definition 10.61 (Comparison atlas). *The atlas used below is the maximal smooth atlas of the continuation orbits in $P_{\text{reg}}^{\text{geom}}$. It is derived from the APF differential space and its admissible-flow orbits, not assumed on the full reduced carrier.*

10.1 What finite-ledger assumptions become geometrically

The finite-ledger assumptions now divide cleanly. Closed-world completeness provides point separation and complete readout fibres. Finite instruction provides second countability. Lossless postprocessing preserves those fibres, and smooth saturation supplies the canonical Sikorski functional calculus. Finite capacity provides finite-channel sufficiency; smooth local factorization is the separate, explicitly named finite-differential-sufficiency gate. Admissible continuation then supplies a family complete in the orbit-theorem sense of flow-generating derivations, whose orbits are immersed smooth manifolds. Constant rank supplies the quotient bundle used by the response geometry.

10.2 Failure witnesses

The differential-space formulation makes failure informative:

- a Stone-like zero-dimensional sector has records but no continuation orbit of positive dimension;
- a rank jump is an intrinsic singular stratum;
- failure of finite-channel sufficiency signals saturation or unbounded local capacity;
- failure of finite differential sufficiency signals nonsmooth local factorization even when the physical record remains complete;
- a jump in the common derivation/cotangent dimension is a singularity witness;
- failure of concrete-model separation means only the numerical model, not the universal physical response, is incomplete.

11. Positive cost geometry and Lorentzian comparison geometry

The pullback construction above is positive. Physics also requires a signed causal comparison structure. These are related structures, but they are not the same tensor obtained under different notation.

Proposition 11.1 (Positive pullbacks cannot produce Lorentzian signature). *Let h be a positive semidefinite bilinear form on a process tangent space and $B : T_x\mathcal{M} \rightarrow T_N\mathcal{P}$ linear. Then B^*hB is positive semidefinite. In particular, no construction of the form T^*T with a positive ambient inner product can have Lorentzian index one.*

Proof. For every u , $(B^*hB)(u, u) = h(Bu, Bu) \geq 0$. A Lorentzian form has vectors of both signs, so it cannot arise in this way. $\square$

Principle 11.2 (Causal cone from finite coordination). *If comparison continuations have finite maximum coordination throughput, tangent directions separate into causally admissible, null-boundary, and unreachable classes. A smooth represented regime should encode these classes by a cone field $C_x \subset T_x\mathcal{M}$.*

Principle 11.3 (Temporal orientation from irreversible record order). *Papers 1, 3, and 37 supply the distinction between coherent held identity and irreversible record-bearing order. Only after a causal cone field has been independently constructed may Paper 9 represent that record preorder as a time orientation of the cone. Record irreversibility therefore selects between the two components of an already available causal cone; it does not by itself derive Lorentzian signature or the cone.*

Gate 11.4 (Signed Lorentz–Finsler comparison gate). *A signed comparison representation is available when there is a continuous two-homogeneous function*

$$Q_x^{\text{cmp}} : T_x \mathcal{M} \longrightarrow \mathbb{R}$$

that is C^2 on the relevant nonzero conic domains, has null set equal to the boundary of the finite-coordination cone, and whose vertical Hessian

$$g_{x,v}^{\text{cmp}} := \frac{1}{2} D_v^2 Q_x^{\text{cmp}}$$

has Lorentzian index one. If this Hessian is independent of v , then

$$Q_x^{\text{cmp}}(v) = q_x(v, v)$$

for a Lorentzian bilinear form q_x .

Gate 11.5 (Cost–causal bridge). *The positive operational cost gauge c_x^{anc} and the signed comparison function Q_x^{cmp} must be related by an explicit physical dictionary. Possibilities include a common normalization on selected timelike or spacelike probes, a bound between implementation cost and $\sqrt{|Q^{\text{cmp}}|}$, or a variational relation. No universal identity is assumed, especially on null directions.*

Boundary 11.6 (Why sectorwise polarization is insufficient). Timelike and spacelike cones are not linear subspaces. Imposing a parallelogram identity separately on positive- and negative-sign sectors does not by itself reconstruct the cross-sector bilinear pairing or a global Lorentzian form. The signed two-homogeneous function or an equivalent cone Lagrangian must be defined on a common tangent domain before its Hessian can certify signature.

Gate 11.7 (Dimension gate). *The dimension gate closes when the number of independent local comparison modes is fixed by substrate capacity, representation stability, admissible interaction structure, and the comparison-response anchor. A scalar-register lower bound may constrain one implementation architecture, but it is not by itself a theorem that the substrate tangent bundle has dimension four. This draft continues to treat $(1 + 3)$ as an input to the weak-field application.*

Audit Note 11.8 (Two geometric tensors). In the quadratic branch the paper may carry both

$$g_x^{\text{cost}} \quad \text{and} \quad q_x^{\text{cmp}}.$$

The first is positive and measures calibrated response effort. The second is signed and encodes causal comparison. They may share preferred directions or be linked by the recruitment dynamics, but identifying them requires a theorem beyond the pullback construction.

12. The geometric recruitment functional

The geometric substrate is not a passive manifold. It is the substrate that carries comparison continuations. Matter and stress-energy distinctions recruit that substrate; the recruitment load changes the future cost of comparison. The functional below is the geometric specialization of the recruitment cost functional of Papers 24/25.

Definition 12.1 (Effective geometric recruitment load). *Let $\chi_{\text{rec}}(x)$ denote the effective accumulated recruitment load carried by the geometric substrate at x . The notation χ , rather than ρ , is deliberate: the leading exterior point-mass profile is potential-like, scaling as $1/r$, not as a volumetric density.*

Definition 12.2 (Quadratic geometric recruitment functional). *In the linear-response regime around an unloaded substrate, a scalar recruitment load field χ sourced by a mass-energy distinction $\mathcal{J}$ is represented by*

$$E_{\text{rec}}[\chi] = \frac{1}{2} \int_{\mathcal{M}} \mathcal{K}^{ij} \partial_i \chi \partial_j \chi d^3x - \int_{\mathcal{M}} \mathcal{J} \chi d^3x + O(\chi^3),$$

where $\mathcal{K}^{ij}$ is the unloaded spatial comparison stiffness. The cubic and higher terms encode nonlinear self-coupling and are not fixed in this draft.

Assumption 12.3 (Isotropic exterior kernel). *For a static, isolated, spherically symmetric source in the exterior linear-response regime, the unloaded spatial comparison stiffness is isotropic:*

$$\mathcal{K}^{ij} = K_s \delta^{ij}.$$

Proposition 12.4 (Exterior point-source profile in the linear-response regime). *Under the quadratic recruitment functional and isotropic exterior kernel, a point source $\mathcal{J}(x) = \mu M \delta^{(3)}(x)$ has an exterior equilibrium profile*

$$\chi_{\text{rec}}(r) = \frac{\mu M}{4\pi K_s r}$$

up to the normalization convention for μ .

Proof. Varying E_{rec} gives the Euler-Lagrange equation

$$-\partial_i(\mathcal{K}^{ij} \partial_j \chi) = \mathcal{J}.$$

With $\mathcal{K}^{ij} = K_s \delta^{ij}$, this becomes

$$-K_s \nabla^2 \chi = \mu M \delta^{(3)}(x).$$

Using $\nabla^2(1/r) = -4\pi \delta^{(3)}(x)$, the exterior solution is

$$\chi_{\text{rec}}(r) = \frac{\mu M}{4\pi K_s r},$$

with any additive constant fixed by the boundary condition $\chi \rightarrow 0$ as $r \rightarrow \infty$. $\square$

Remark 12.5 (What this does and does not close). Proposition 12.4 closes only the Green's-function step once the quadratic isotropic kernel is assumed. It does not derive the kernel from APF primitives, does not derive the source normalization μ , and does not derive Newton's constant. It is nevertheless useful: it shows that the required $1/r$ exterior profile is the natural equilibrium profile of the simplest geometric recruitment functional.

13. Cost-curvature response

Principle 13.1 (Linear cost-curvature response). *In the exterior weak-load regime, the time-comparison cost-curvature responds linearly to the effective accumulated geometric recruitment load:*

$$K_t(x) = K_{t0}(1 - \alpha \chi_{\text{rec}}(x)),$$

where K_{t0} is the unloaded time-comparison stiffness and α is the coupling between committed load and residual time-comparison stiffness.

The sign is physical: substrate already committed to carrying recruitment load has reduced residual stiffness for additional identity-carrying time comparisons. In represented geometry, this reduced stiffness appears as slower local proper-time accumulation relative to the unloaded coordinate normalization. In substrate-position-axis terms (Paper 0 §sec:substrate_position_axis), the sign is the substrate's middle-regime saturation-direction physics at the geometric-substrate scale: the closer the substrate is to its saturation regime — the more committed its capacity is to existing recruitment load — the less residual middle-regime capacity remains for additional comparison continuations. The Bekenstein-bound saturation-regime cardinal datum (Paper 6 Supplement v2.10 §sec:Lambda_subspace, $S_{\text{dS}} = 61 \log 102$) is the global-scale saturation limit of the same response.

Boundary 13.2 (Status of the response law). The response law is a structural commitment in this draft. A closed derivation would require a nonlinear geometric recruitment functional, expansion around the loaded equilibrium, and proof that the leading correction to K_{t0} is exactly $-\alpha\chi_{\text{rec}}$. That is one of the main open tasks of this paper.

13.1 Minimal perturbative response model

The following lemma does not close the response law from APF primitives. It formalizes the exact kind of load-probe coupling that would make the law unavoidable in a weak-load effective theory.

Definition 13.3 (Time-comparison probe). *Let u be a small normalized probe field representing an infinitesimal rest-frame identity-comparison tick. In the unloaded substrate its leading quadratic cost is*

$$E_t[u] = \frac{1}{2} \int K_{t0} u^2 d\lambda,$$

where $d\lambda$ is the local comparison-volume element appropriate to the clock sector.

Assumption 13.4 (Bilinear load-probe coupling). *In the weak-load regime, the leading coupling between accumulated geometric recruitment load χ and the time-comparison probe u is*

$$E_t[u; \chi] = \frac{1}{2} \int (K_{t0} - \lambda\chi(x)) u(x)^2 d\lambda + O(u^3, \chi^2 u^2),$$

with $\lambda > 0$. The minus sign states residual-stiffness depletion: already committed geometric load reduces available stiffness for an additional identity-comparison tick.

Proposition 13.5 (Linear response from bilinear load-probe coupling). *Under the bilinear load-probe coupling, the local time-comparison stiffness at fixed load is*

$$K_t(x) = K_{t0} - \lambda\chi(x) + O(\chi^2),$$

or equivalently

$$K_t(x) = K_{t0}(1 - \alpha\chi(x)) + O(\chi^2), \quad \alpha := \lambda/K_{t0}.$$

Proof. The local stiffness is the second variation of the probe cost with respect to the probe amplitude at the loaded equilibrium:

$$K_t(x) = \left. \frac{\delta^2 E_t[u; \chi]}{\delta u(x)^2} \right|_{u=0}.$$

Taking the second variation of the displayed effective cost gives $K_{t0} - \lambda\chi(x)$ to first order in χ . Dividing by K_{t0} gives the normalized coefficient $\alpha = \lambda/K_{t0}$. $\square$

Audit Note 13.6 (What the response lemma buys). Proposition 13.5 converts the response law from a naked ansatz into a precise perturbative target. The remaining open work is now sharper: derive the negative bilinear load-probe coupling from the nonlinear geometric substrate functional, rather than merely assert the final linear response.

Definition 13.7 (Newtonian calibration). *In physical units, define the exterior point-mass calibration by*

$$\alpha\chi_{\text{rec}}(r) = \frac{2GM}{rc^2}.$$

Equivalently, the combination $\alpha\mu/(4\pi K_s)$ is fixed to $2G/c^2$. In this draft, G is a substrate-specific dimensional input, not an internally derived number.

Corollary 13.8 (Schwarzschild time factor as calibrated cost curvature). *Under the point-source profile and Newtonian calibration,*

$$K_t(r) = K_{t0} \left(1 - \frac{2GM}{rc^2} \right)$$

in the exterior linear-response regime.

14. Proper time as normalized signed clock-sector comparison length

Definition 14.1 (Proper-time dictionary). *In the represented metric branch, the proper-time element for a static identity-preserving held continuation is the normalized signed clock-sector comparison length assigned by the represented causal kernel:*

$$d\tau(r) = \sqrt{K_t(r)/K_{t0}} dt.$$

This dictionary treats dt as the unloaded coordinate comparison parameter and $d\tau$ as the local identity-continuation clock tick.

Remark 14.2 (Why the square root). The local line element is quadratic in infinitesimal comparison displacement. Cost-curvature multiplies the quadratic form; the represented cost length is therefore the square root of the curvature-normalized quadratic element. This is a representation dictionary, not yet a derivation from a fully closed substrate action.

Boundary 14.3 (Rest-frame bridge, not a signature derivation). The proper-time dictionary is a calibration of the signed comparison kernel on static timelike probes. It does not follow merely from the positive pullback response norm, and it does not identify general implementation cost with Lorentzian interval. Null directions can have zero signed interval while carrying nonzero physical maintenance or realization cost.

Definition 14.4 (Clock sector). *The clock sector of a represented geometry is the part of the comparison-cost kernel that controls static rest-frame identity continuations. In a static coordinate representation it is encoded by a lapse factor*

$$N(x) := \frac{d\tau(x)}{dt}.$$

The redshift calculation constrains this sector only.

Definition 14.5 (Spatial comparison sector). *The spatial comparison sector is the part of the comparison-cost kernel that controls endpoint comparison at fixed clock label. In a weak-field isotropic representation one may write, schematically,*

$$ds^2 = -N(x)^2 c^2 dt^2 + S(x)^2 d\mathbf{x}^2,$$

where redshift fixes the lapse factor N but not the spatial scale factor S .

Proposition 14.6 (Redshift does not fix full geometry). *Any two static represented geometries with the same lapse factor $N(r) = d\tau(r)/dt$ give the same static gravitational redshift between radii, even if their spatial comparison sectors differ. Therefore a redshift recovery alone cannot establish full GR.*

Proof. For static emitter and receiver, local frequency is inverse local proper period. The same coordinate period is compared against the two local lapse factors, so

$$\frac{\omega_A}{\omega_B} = \frac{N(r_B)}{N(r_A)}.$$

No spatial comparison factor enters this expression. Hence all represented metrics with the same lapse produce the same redshift. $\square$

Proposition 14.7 (Leading-order redshift recovery from cost-curvature response). *Consider a static, spherically symmetric recruitment load generated by a point mass M , in the exterior linear-response regime. Suppose*

$$K_t(r) = K_{t0}(1 - \alpha\chi_{\text{rec}}(r)), \quad \alpha\chi_{\text{rec}}(r) = \frac{2GM}{rc^2},$$

and suppose proper time is represented by normalized identity-continuation cost:

$$d\tau(r) = \sqrt{K_t(r)/K_{t0}} dt.$$

Then

$$d\tau(r) = \sqrt{1 - \frac{2GM}{rc^2}} dt.$$

Consequently, for a photon emitted by a static observer at r_B and received by a static observer at $r_A > r_B$,

$$\frac{\omega_A}{\omega_B} = \sqrt{\frac{1 - 2GM/r_Bc^2}{1 - 2GM/r_Ac^2}}.$$

In the weak-field limit,

$$\frac{\omega_A}{\omega_B} \approx 1 - \frac{GM}{c^2} \left(\frac{1}{r_B} - \frac{1}{r_A} \right).$$

Proof. The first displayed formula follows immediately from substituting the calibrated recruitment load into the cost-curvature response law and then applying the proper-time dictionary. For a static emitter and receiver, the same coordinate-period label is compared against two different local proper-time rates. Local frequency is inverse local proper period, hence

$$\omega_A = \omega_B \frac{d\tau(r_B)}{d\tau(r_A)}.$$

Substitution gives the redshift ratio. Expanding $\sqrt{1 - 2GM/rc^2}$ to first order gives the weak-field expression. $\square$

Boundary 14.8 (Boundary of the result). Proposition 14.7 is a conditional recovery, not a closed derivation of GR. It assumes: (i) the linear cost-curvature response, (ii) the exterior point-source recruitment profile, (iii) the proper-time-as-normalized-cost dictionary, and (iv) G as the substrate coupling in physical units. Deriving these from substrate primitives, extending the result beyond the exterior linear-response regime, and recovering the full Einstein equations are open.

15. No-smuggling audit for the redshift recovery

The redshift recovery is useful only if its inputs are named without disguise. The calculation has four layers.

Layer	Status	Content
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Continuation grammar	Imported	The judgment $\Gamma \Vdash_C D \rightsquigarrow E$ and the interpretation of geometry as comparison-continuation cost are imported from Paper 10.
Exterior profile	Cond./proved	The $1/r$ profile is proved here only after assuming an isotropic quadratic exterior kernel.
Clock response	Assumed	$K_t = K_{t0}(1 - \alpha\chi)$ is a linear-response commitment. Its sign and linearity are physically motivated but not derived from a nonlinear substrate action.
Unit calibration	Input	The equality $\alpha\chi = 2GM/rc^2$ imports the physical coupling G . The calculation recovers the form of the redshift, not the numerical value of G .
Proper-time dictionary	Representational	$d\tau = \sqrt{K_t/K_{t0}}dt$ is the metric-regime dictionary between normalized identity-continuation cost and clock time.

Audit Note 15.1 (What is actually earned). The calculation earns the following claim: if a point mass produces a potential-like recruitment load, if time-comparison stiffness is reduced linearly by that load, and if proper time is normalized cost length, then the static clock-redshift sector has the Schwarzschild form at leading exterior order. It does not earn the spatial metric, the equivalence principle, nonlinear field equations, or a derived value of G .

Audit Note 15.2 (What would close the result). To turn the recovery into a derivation, one must derive the isotropic Green's-function kernel, the universal response law, the proper-time dictionary, and the physical coupling normalization from APF substrate primitives. These are not cosmetic gaps; they are the main mathematical work of Paper 9.

16. Newtonian acceleration as a cost-gradient effect

The redshift calculation recovers the clock sector. The next target is motion. In represented weak-field geometry, slow test bodies accelerate along gradients of the same potential that controls g_{tt} . Cost-theoretically, this should emerge because least-cost identity-carrying worldlines bend toward lower continuation cost.

Definition 16.1 (Effective cost potential). *Define the effective cost potential Φ_C by*

$$\frac{K_t(r)}{K_{t0}} = 1 + \frac{2\Phi_C(r)}{c^2}.$$

For the calibrated point-source profile,

$$\Phi_C(r) = -\frac{GM}{r}.$$

Proposition 16.2 (Newtonian acceleration under the geodesic-cost dictionary). *Assume the metric-representation gate closes and slow test-body worldlines extremize normalized identity-continuation cost. In the weak-field, slow-velocity regime with*

$$\frac{K_t}{K_{t0}} = 1 + \frac{2\Phi_C}{c^2},$$

the represented coordinate acceleration is

$$\mathbf{a} = -\nabla\Phi_C.$$

For $\Phi_C = -GM/r$, this gives the inverse-square law.

Proof sketch. Under the geodesic-cost dictionary, the action for a slow test body is proportional to

$$\int d\tau \approx \int \left(1 + \frac{\Phi_C}{c^2} - \frac{v^2}{2c^2} \right) dt$$

up to an irrelevant constant normalization and sign convention. Extremizing is equivalent to extremizing the ordinary weak-field Lagrangian $L = v^2/2 - \Phi_C$, yielding $\mathbf{a} = -\nabla\Phi_C$. The step is conditional on the metric and geodesic-cost dictionaries. $\square$

Remark 16.3 (Why this matters). The same scalar cost load controls both clock rate and slow-body acceleration. This is the beginning of an equivalence-principle-like universality claim, but it is not yet a derivation of the equivalence principle. To close that claim, the response law must be shown universal for all admissible rest-frame identity continuations, independent of matter composition.

17. Weak-field two-sector kernel and the missing spatial response

The clock-sector recovery should be tested against the standard weak-field hierarchy. The redshift calculation gives a lapse. A gravitational theory also needs the spatial comparison response. The next formal object is therefore not a scalar clock stiffness alone, but a two-sector normalized cost kernel.

Definition 17.1 (Normalized two-sector weak-field cost kernel). *In a static, isotropic, weak-field represented regime, write the normalized comparison-cost line element as*

$$ds_C^2 = -\Theta_t(r)c^2dt^2 + \Theta_s(r)d\mathbf{x}^2 + O(c^{-4}),$$

where

$$\Theta_t(r) := \frac{K_t(r)}{K_{t0}}, \quad \Theta_s(r) := \frac{K_s(r)}{K_{s0}}.$$

The redshift result fixes Θ_t . It does not fix Θ_s .

Definition 17.2 (Weak-field cost potential). *For a static exterior point source, define the cost potential Φ_C by*

$$\Theta_t(r) = 1 + \frac{2\Phi_C(r)}{c^2} + O(c^{-4}).$$

Under the Newtonian calibration of the scalar recovery,

$$\Phi_C(r) = -\frac{GM}{r}.$$

Definition 17.3 (Spatial response parameter). *The weak-field spatial comparison response is defined by*

$$\Theta_s(r) = 1 - 2\gamma_C \frac{\Phi_C(r)}{c^2} + O(c^{-4}).$$

The dimensionless coefficient γ_C is the APF/cost-theoretic counterpart of the standard PPN parameter γ_{PPN} . General relativity has $\gamma_C = 1$.

Proposition 17.4 (Clock observables fix only the lapse). *Gravitational time dilation, static redshift, and Newtonian slow-body acceleration determine the leading clock potential Φ_C but do not determine the spatial-response coefficient γ_C .*

Proof. The redshift and time-dilation formulae depend only on the lapse factor $\Theta_t^{1/2}$. The Newtonian slow-motion limit follows from the same lapse through the weak-field particle action. The spatial factor Θ_s enters null propagation and spatial-curvature observables at the same post-Newtonian order, but it does not enter the static redshift ratio and does not affect the lowest-order slow-velocity acceleration law. Therefore the clock sector fixes Φ_C and leaves γ_C free. $\square$

Gate 17.5 (Spatial-response / PPN- γ gate). *To recover the full leading weak-field Schwarzschild sector, the geometric substrate must determine the spatial comparison response*

$$\Theta_s(r) = 1 - 2\frac{\Phi_C(r)}{c^2} + O(c^{-4}),$$

that is,

$$\gamma_C = \gamma_{\text{PPN}} = 1.$$

Until this gate closes, the paper has recovered the clock sector and Newtonian limit, but not light bending, Shapiro delay, or the full first-post-Newtonian exterior metric.

Gate 17.6 (Reciprocal clock-ruler response). *The spatial-response gate can be sharpened into a candidate closure principle. In the exterior weak-field scalar-load regime, the time-comparison and spatial-comparison sectors are required to be reciprocal to first post-Newtonian order:*

$$\Theta_s(r)\Theta_t(r) = 1 + O(c^{-4}),$$

or equivalently

$$\Theta_s(r) = \Theta_t(r)^{-1} + O(c^{-4}).$$

Physical reading: clock comparison and ruler comparison are conjugate faces of the same finite-coordination substrate. A load that reduces the residual rate of identity-carrying clock continuation must increase the cost of spatial endpoint comparison by the inverse factor if the same local null coordination boundary is to be used by all admissible probes. This is a gate, not a theorem: it must ultimately be derived from the microscopic comparison functional.

Proposition 17.7 (Reciprocal response implies the PPN spatial coefficient). *Assume the two-sector weak-field form*

$$\Theta_t(r) = 1 + \frac{2\Phi_C(r)}{c^2} + O(c^{-4}), \quad \Theta_s(r) = 1 - 2\gamma_C \frac{\Phi_C(r)}{c^2} + O(c^{-4}).$$

If the reciprocal clock-ruler gate holds, then

$$\gamma_C = 1.$$

Proof. Multiplying the two sector factors gives

$$\begin{aligned} \Theta_s\Theta_t &= \left(1 - 2\gamma_C \frac{\Phi_C}{c^2}\right) \left(1 + 2\frac{\Phi_C}{c^2}\right) + O(c^{-4}) \\ &= 1 + 2(1 - \gamma_C) \frac{\Phi_C}{c^2} + O(c^{-4}). \end{aligned}$$

The reciprocal gate requires this product to be unity through order c^{-2} . For a nonzero exterior potential this forces $1 - \gamma_C = 0$, hence $\gamma_C = 1$. $\square$

Audit Note 17.8 (Why naive same-sign scalar depletion fails). A tempting first guess is that every comparison stiffness is depleted in the same scalar way:

$$\Theta_s(r) = \Theta_t(r) = 1 + \frac{2\Phi_C(r)}{c^2} + O(c^{-4}).$$

Compared with the PPN form $\Theta_s = 1 - 2\gamma_C\Phi_C/c^2$, this gives $\gamma_C = -1$. Thus redshift would still be recovered from the clock sector, but the spatial-curvature contribution needed for the standard two-sector weak-field observables would have the wrong coefficient. The geometric substrate cannot be modeled as uniform same-sign depletion of every comparison channel. The spatial sector must be conjugate, tensorial, or otherwise constrained; this is precisely the content of Gate 17.6 and Gate 17.5.

17.1 Candidate derivation route: clock-ruler comparison-cell conservation

The reciprocal clock-ruler condition should not remain a slogan. The following definitions isolate a sharper operational certificate. The certificate is still a gate, but it is weaker and more testable than simply assuming $\gamma_C = 1$: it says that, direction by direction, the elementary comparison cell built from one clock tick and one spatial endpoint ruler is preserved by a static scalar load to first post-Newtonian order.

Definition 17.9 (Directionwise clock-ruler comparison cell). *Fix a represented spatial comparison direction e in the static weak-field regime. The normalized clock-ruler comparison cell in that direction is*

$$\mathcal{A}_e(r) := \left(\frac{d\tau(r)}{dt} \right) \left(\frac{dl_e(r)}{dx_e} \right) = \sqrt{\Theta_t(r)\Theta_{s,e}(r)},$$

where $dl_e = \sqrt{\Theta_{s,e}} dx_e$ is the represented spatial comparison length in direction e . In the isotropic weak-field sector, $\Theta_{s,e} = \Theta_s$ for all spatial directions.

Gate 17.10 (Directionwise comparison-cell conservation). *In an exterior, static, scalar-load regime with no shear, torsion, anisotropic substrate recruitment, or composition-dependent comparison channel, the elementary clock-ruler comparison cell is conserved to first post-Newtonian order:*

$$\mathcal{A}_e(r) = 1 + O(c^{-4})$$

for every represented spatial direction e .

Physical reading: a scalar geometric load may redistribute comparison capacity between clock-rate continuation and endpoint-ruler continuation, but it does not create or destroy the local directionwise comparison cell. Clock slowing and ruler stretching are conjugate responses of the same finite-coordination substrate.

Proposition 17.11 (Cell conservation implies reciprocal clock-ruler response). *In the static isotropic two-sector weak-field kernel, directionwise comparison-cell conservation implies*

$$\Theta_s(r)\Theta_t(r) = 1 + O(c^{-4}).$$

Consequently, under the PPN expansion of Proposition 17.7, it implies $\gamma_C = 1$.

Proof. In the isotropic sector, $\mathcal{A}_e = \sqrt{\Theta_t\Theta_s}$ is independent of e . The cell-conservation gate gives $\sqrt{\Theta_t\Theta_s} = 1 + O(c^{-4})$. Squaring gives $\Theta_t\Theta_s = 1 + O(c^{-4})$. The implication $\gamma_C = 1$ then follows from Proposition 17.7. $\square$

Audit Note 17.12 (What the cell-conservation route does not prove). Cell conservation is not the local speed of light, coordinate invariance, or redshift in disguise. Local null measurements give a universal measured speed once the metric representation is assumed; they do not by themselves fix Θ_s . The new content is the claim that a static scalar recruitment load preserves the directionwise clock-ruler comparison cell at order c^{-2} . This can fail if the load creates anisotropic shear, if clock and ruler comparisons draw on independent substrate reservoirs, if matter species couple to different comparison kernels, or if the geometric load changes the local comparison-cell budget rather than redistributing it. These are empirical and mathematical failure modes, not notational details.

17.2 Sharper candidate route: canonical comparison pairing

The cell-conservation gate can itself be decomposed. A comparison between events has two faces: a primal clock-ruler displacement and a dual frequency-wavevector probe. The dimensionless action/phase pairing between these faces is the invariant bookkeeping object. This subsection isolates the extra condition needed to make that pairing imply reciprocal clock-ruler response.

Definition 17.13 (Local canonical comparison pair). *Fix a spatial comparison direction e in a represented local frame. The primal comparison variables are*

$$q_e = (\tau, \ell_e),$$

where τ is clock continuation and ℓ_e is endpoint-ruler continuation in direction e . The dual probe variables are

$$p_e = (\omega, k_e),$$

where ω is frequency cost and k_e is directional wave-number cost. The local comparison one-form is

$$\vartheta_e = -\omega d\tau + k_e d\ell_e.$$

Equivalently, $\vartheta_e/\hbar$ is the local dimensionless phase increment of a probe used to test the comparison.

Gate 17.14 (Canonical load transport). *A static scalar recruitment load acts canonically on the local comparison pair if, direction by direction, its transport map preserves the comparison one-form up to terms of order c^{-4} :*

$$T_\chi^* \vartheta_e = \vartheta_e + O(c^{-4}).$$

For a diagonal weak-field response this means that if

$$d\tau \mapsto a_t d\tau, \quad d\ell_e \mapsto a_s d\ell_e,$$

then the dual variables transform inversely,

$$\omega \mapsto a_t^{-1} \omega, \quad k_e \mapsto a_s^{-1} k_e,$$

so that the phase/action pairing is not changed by relabeling the same comparison through the loaded substrate.

Audit Note 17.15 (Canonical pairing alone is not enough). Gate 17.14 preserves the action/phase pairing, but it does not by itself force $a_t a_s = 1$. Any nonzero product $a_t a_s$ can be paired with inverse dual scaling and still preserve ϑ_e . Therefore the spatial coefficient $\gamma_C = 1$ requires one additional physical statement: the scalar load may exchange clock and ruler comparison capacity, but it may not dilate the elementary primal clock-ruler comparison cell at order c^{-2} .

Gate 17.16 (Trace-free clock-ruler load transport). *In the exterior static scalar-load regime, after removing coordinate relabelings and anisotropic shear, the primal action of the load on each local clock-ruler comparison plane is trace-free to first post-Newtonian order. Equivalently, there exists a scalar load parameter $\sigma(r) = O(c^{-2})$ such that*

$$d\tau \mapsto e^{\sigma(r)} d\tau, \quad d\ell_e \mapsto e^{-\sigma(r)} d\ell_e$$

for every spatial direction e , up to $O(c^{-4})$.

Physical reading: a static scalar geometric load does not create an independent local comparison-cell resource. It transfers comparison capacity between conjugate clock and ruler channels. The trace part would represent creation or destruction of the directionwise comparison cell; the trace-free part represents redistribution.

Proposition 17.17 (Trace-free transport implies cell conservation). *Assume the represented two-sector weak-field kernel and trace-free clock-ruler load transport. Then directionwise clock-ruler comparison-cell conservation holds:*

$$\mathcal{A}_e(r) = 1 + O(c^{-4}).$$

Consequently,

$$\Theta_s(r)\Theta_t(r) = 1 + O(c^{-4}) \quad \text{and} \quad \gamma_C = 1.$$

Proof. By definition, the normalized clock factor is $d\tau/dt = \sqrt{\Theta_t}$, and the normalized ruler factor in direction e is $d\ell_e/dx_e = \sqrt{\Theta_{s,e}}$. Trace-free transport gives these factors as e^σ and $e^{-\sigma}$, up to $O(c^{-4})$. Hence

$$\mathcal{A}_e = \left(\frac{d\tau}{dt}\right) \left(\frac{d\ell_e}{dx_e}\right) = e^\sigma e^{-\sigma} + O(c^{-4}) = 1 + O(c^{-4}).$$

The remaining implications are exactly Propositions 17.7 and 17.11. $\square$

Corollary 17.18 (Nested spatial-gate chain). *The leading spatial coefficient can be closed by any of the following increasingly primitive certificates:*

$$\begin{aligned} \text{trace-free clock-ruler load transport} &\implies \text{directionwise cell conservation} \\ &\implies \Theta_s \Theta_t = 1 + O(c^{-4}) \\ &\implies \gamma_C = 1. \end{aligned}$$

Thus the next mathematical target is sharper than $\gamma_C = 1$: derive trace-free transport from the geometric recruitment functional, or exhibit a countermodel showing that APF permits a trace component.

17.3 Variational route: fixed-cell response and the trace obstruction

Trace-free transport should itself be generated by a local response principle, not merely imposed. The cleanest way to state the target is to decompose the diagonal clock-ruler load response into its trace and trace-free parts.

Definition 17.19 (Logarithmic clock-ruler response variables). *Fix a represented spatial direction e . Write the local diagonal load response as*

$$\frac{d\tau}{dt} = a_t = e^{\xi_t}, \quad \frac{d\ell_e}{dx_e} = a_s = e^{\xi_s}.$$

Define the trace and exchange variables

$$u_e := \xi_t + \xi_s, \quad v_e := \xi_t - \xi_s.$$

Then $u_e = \log \mathcal{A}_e$ is the logarithmic clock-ruler cell dilation, while v_e is the trace-free exchange between clock and ruler comparison channels.

Gate 17.20 (Fixed-cell variational response). *In the exterior static scalar-load regime, after removing coordinate relabelings and shear, the local response of a directionwise clock-ruler comparison plane is governed to quadratic order by an effective potential of the form*

$$\mathcal{V}_e(u_e, v_e; \chi) = \frac{1}{2} A_e u_e^2 + \frac{1}{2} B_e (v_e - \beta_e \chi)^2 + O(|u_e|^3 + |v_e|^3 + \chi^2 |u_e| + \chi^2 |v_e|),$$

with $A_e, B_e > 0$. The scalar load sources the exchange mode v_e but does not source the trace mode u_e at first order.

Physical reading: committed geometric recruitment may shift capacity between clock continuation and ruler comparison, but a scalar static load has no independent channel for creating or destroying the elementary clock-ruler comparison cell. Any first-order source term proportional to χu_e would be a new local comparison-resource reservoir and must be separately justified.

Proposition 17.21 (Fixed-cell variational response implies trace-free transport). *Assume the fixed-cell variational response gate and $\chi = O(c^{-2})$. The local equilibrium response satisfies*

$$u_e = O(\chi^2) = O(c^{-4}), \quad v_e = \beta_e \chi + O(\chi^2).$$

Consequently

$$\mathcal{A}_e = e^{u_e} = 1 + O(c^{-4}),$$

so directionwise comparison-cell conservation holds. In the isotropic two-sector weak-field regime this implies

$$\Theta_s \Theta_t = 1 + O(c^{-4}), \quad \gamma_C = 1.$$

Proof. Equilibrium requires $\partial \mathcal{V}_e / \partial u_e = 0$ and $\partial \mathcal{V}_e / \partial v_e = 0$. To first order,

$$\frac{\partial \mathcal{V}_e}{\partial u_e} = A_e u_e + O(\chi^2), \quad \frac{\partial \mathcal{V}_e}{\partial v_e} = B_e (v_e - \beta_e \chi) + O(\chi^2).$$

Since $A_e, B_e > 0$, the equilibrium solution is

$$u_e = O(\chi^2), \quad v_e = \beta_e \chi + O(\chi^2).$$

Because $\chi = O(c^{-2})$, the trace is absent through order c^{-2} . Hence $\mathcal{A}_e = e^{u_e} = 1 + O(c^{-4})$. The final implications are Propositions 17.7 and 17.11. $\square$

17.4 Operator form of the no-trace target

The two-variable potential above is the scalar normal form of a more general loaded-equilibrium problem. In the represented comparison-metric regime, let q_0 be the unloaded local comparison kernel and let $\mathcal{S}$ be the finite-dimensional gauge-fixed space of symmetric perturbations h at that point. Write

$$\text{tr}_{q_0} h := q_0^{ab} h_{ab}, \quad P_{\text{tr}} h := \frac{1}{m} (\text{tr}_{q_0} h) q_0, \quad P_{\text{TF}} := I - P_{\text{tr}},$$

where $m = \dim \mathcal{M}$ in the represented regime.

Assumption 17.22 (Gauge-fixed quadratic recruitment response). *Around q_0 , the local recruitment functional has expansion*

$$\begin{aligned}\mathcal{E}_{\text{rec}}[q_0 + h, \chi] &= \mathcal{E}_0 + \frac{1}{2}\langle h, \mathcal{K}h \rangle - \chi\langle \mathcal{J}, h \rangle \\ &\quad + O(\|h\|^3 + \chi\|h\|^2 + \chi^2\|h\|),\end{aligned}$$

where $\mathcal{K} : \mathcal{S} \rightarrow \mathcal{S}$ is self-adjoint and invertible after coordinate-gauge zero modes are removed, and $\mathcal{J} \in \mathcal{S}$ is the scalar-load source.

Theorem 17.23 (Linear response and the exact no-trace operator). *Under Assumption 17.22, the equilibrium perturbation is*

$$h_\chi = \chi \mathcal{K}^{-1} \mathcal{J} + O(\chi^2).$$

Its first-order comparison-volume variation is

$$\delta_\chi \log \sqrt{|\det(q_0 + h_\chi)|} = \frac{\chi}{2} \text{tr}_{q_0}(\mathcal{K}^{-1} \mathcal{J}) + O(\chi^2).$$

Consequently the no-trace gate is exactly

$$\boxed{P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0}.$$

Proof. The stationary equation is

$$\mathcal{K}h_\chi - \chi \mathcal{J} + O(\|h_\chi\|^2 + \chi\|h_\chi\| + \chi^2) = 0.$$

Invertibility and the implicit-function expansion give $h_\chi = \chi \mathcal{K}^{-1} \mathcal{J} + O(\chi^2)$. The determinant variation formula gives

$$\log \sqrt{|\det(q_0 + h)|} = \log \sqrt{|\det q_0|} + \frac{1}{2} \text{tr}_{q_0} h + O(\|h\|^2).$$

Substitution proves the volume formula. Vanishing of the trace component is equivalent to $P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0$. $\square$

Corollary 17.24 (Source-selection route). *If*

$$P_{\text{tr}} \mathcal{J} = 0$$

and $\mathcal{K}$ preserves the trace-free subspace, equivalently

$$P_{\text{tr}} \mathcal{K} P_{\text{TF}} = 0 = P_{\text{TF}} \mathcal{K} P_{\text{tr}},$$

then $P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0$ and the linear response is trace-free.

Corollary 17.25 (Constrained fixed-volume route). *If the admissible variation space is constrained by*

$$\text{tr}_{q_0} h = 0$$

through a Lagrange multiplier or a primitive fixed-comparison-capacity condition, then the equilibrium response is trace-free by construction. The remaining physical obligation is to derive the constraint from admissibility rather than impose it as a coordinate convention.

Audit Note 17.26 (Relation to the scalar potential). In a clock–ruler plane, u_e is the trace coordinate and v_e the trace-free exchange coordinate. The condition that no term χu_e appear is the two-dimensional source-selection statement $P_{\text{tr}}\mathcal{J} = 0$. The stronger invariant target is $P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$, because a trace-free source can still generate trace if the Hessian mixes the trace and trace-free sectors.

Definition 17.27 (Trace-obstruction coefficient). *If the local response potential contains a first-order trace source,*

$$\mathcal{V}_e(u_e, v_e; \chi) = \frac{1}{2}A_e u_e^2 + \frac{1}{2}B_e(v_e - \beta_e \chi)^2 - \lambda_e \chi u_e + O(3),$$

then the equilibrium trace is

$$u_e = \eta_e \chi + O(\chi^2), \quad \eta_e := \lambda_e / A_e.$$

The coefficient η_e is the first-order trace obstruction.

Proposition 17.28 (Trace obstruction is the PPN- γ obstruction). *In the isotropic two-sector weak-field regime, suppose the logarithmic cell dilation has the expansion*

$$u(r) = \zeta_C \frac{\Phi_C(r)}{c^2} + O(c^{-4}).$$

Then

$$\gamma_C = 1 - \zeta_C.$$

Thus the GR spatial coefficient $\gamma_C = 1$ is equivalent to vanishing first-order trace dilation $\zeta_C = 0$.

Proof. By definition,

$$e^{2u} = \Theta_t \Theta_s.$$

Using the two-sector weak-field expansion,

$$\begin{aligned} \Theta_t \Theta_s &= \left(1 + 2\frac{\Phi_C}{c^2}\right) \left(1 - 2\gamma_C \frac{\Phi_C}{c^2}\right) + O(c^{-4}) \\ &= 1 + 2(1 - \gamma_C) \frac{\Phi_C}{c^2} + O(c^{-4}). \end{aligned}$$

Meanwhile $e^{2u} = 1 + 2\zeta_C \Phi_C / c^2 + O(c^{-4})$. Equating coefficients gives $\zeta_C = 1 - \gamma_C$, hence $\gamma_C = 1 - \zeta_C$. $\square$

Audit Note 17.29 (The remaining obstruction). The spatial gate has now been reduced to a concrete question: does static scalar geometric recruitment source the trace mode u_e of the local clock-ruler response? If not, fixed-cell variational response gives trace-free transport and closes $\gamma_C = 1$. If yes, the sourced trace coefficient ζ_C shifts the PPN spatial coefficient by $\gamma_C = 1 - \zeta_C$. The obstruction is no longer vague. It is the existence of a first-order χu_e term in the effective local response potential.

17.5 Generator form: the obstruction is a $GL(2)$ trace

The variational statement above is useful, but the same obstruction can be stated more invariantly. A weak static load acts locally on each clock-ruler comparison plane by a linear response map. The determinant of this map is the comparison-cell factor. The trace of its infinitesimal generator is therefore exactly the first-order cell-dilation obstruction.

Definition 17.30 (Clock-ruler response map). *Fix a represented spatial direction e . After removing coordinate relabelings and shear, write the loaded clock-ruler increments as*

$$\begin{pmatrix} d\tau \\ d\ell_e \end{pmatrix} = R_e(\hat{\chi}) \begin{pmatrix} d\tau_0 \\ d\ell_{e0} \end{pmatrix}, \quad \hat{\chi}(r) := \frac{\Phi_C(r)}{c^2} = O(c^{-2}),$$

where $R_e(0) = I$ and $R_e(\hat{\chi}) \in GL^+(2, \mathbb{R})$ in the represented smooth regime. Its infinitesimal generator is

$$L_e := \left. \frac{dR_e}{d\hat{\chi}} \right|_0, \quad R_e(\hat{\chi}) = I + \hat{\chi}L_e + O(\hat{\chi}^2).$$

The trace component

$$\theta_e := \text{tr } L_e$$

is the infinitesimal comparison-cell dilation rate. The traceless part $L_e - \frac{1}{2}\theta_e I$ is the exchange/shear part of the clock-ruler response.

Proposition 17.31 (Generator trace equals trace obstruction). *In the diagonal isotropic weak-field sector, the logarithmic cell dilation satisfies*

$$u_e(r) = \theta_e \frac{\Phi_C(r)}{c^2} + O(c^{-4}).$$

Consequently

$$\zeta_C = \theta_e, \quad \gamma_C = 1 - \theta_e.$$

Thus the GR spatial coefficient $\gamma_C = 1$ is equivalent, at this order, to the infinitesimal load generator being traceless:

$$\text{tr } L_e = 0.$$

Proof. The comparison-cell factor is

$$\mathcal{A}_e = \det R_e(\hat{\chi}) = 1 + \hat{\chi} \text{tr } L_e + O(\hat{\chi}^2).$$

Since $u_e = \log \mathcal{A}_e$, we obtain

$$u_e = \hat{\chi} \text{tr } L_e + O(\hat{\chi}^2) = \theta_e \Phi_C / c^2 + O(c^{-4}).$$

Comparison with Proposition 17.28 gives $\zeta_C = \theta_e$ and hence $\gamma_C = 1 - \theta_e$. □

17.6 Trace as comparison-volume variation

The $GL(2)$ trace has an exact geometric interpretation once a represented signed comparison kernel is available on the local clock-ruler plane.

Definition 17.32 (Loaded clock-ruler comparison kernel). *Let $q_{e,0}$ be the unloaded nondegenerate comparison kernel on the clock-ruler plane. Represent the loaded kernel on the unloaded increment space by*

$$q_e(\hat{\chi}) := R_e(\hat{\chi})^* q_{e,0} = R_e(\hat{\chi})^\top q_{e,0} R_e(\hat{\chi}).$$

Theorem 17.33 (Trace-volume identity). *For $R_e(\hat{\chi}) = I + \hat{\chi}L_e + O(\hat{\chi}^2)$,*

$$\boxed{\left. \frac{d}{d\hat{\chi}} \log \sqrt{|\det q_e(\hat{\chi})|} \right|_{\hat{\chi}=0} = \text{tr } L_e}.$$

Thus the infinitesimal generator trace is precisely the first variation of the local comparison-volume element.

Proof. The determinant transforms as

$$\det q_e(\hat{\chi}) = \det(R_e)^2 \det q_{e,0}.$$

Because $R_e(0) = I$ and $R_e \in GL^+(2, \mathbb{R})$ locally,

$$\log \sqrt{|\det q_e|} = \log \det R_e + \log \sqrt{|\det q_{e,0}|}.$$

Differentiating at zero and using $\frac{d}{d\hat{\chi}} \log \det R_e|_0 = \text{tr } L_e$ proves the identity. $\square$

Corollary 17.34 (The PPN obstruction is the scalar volume mode). *After clock calibration,*

$$\left. \frac{d}{d\hat{\chi}} \log \sqrt{|\det q_e(\hat{\chi})|} \right|_0 = 1 - \gamma_C.$$

Hence $\gamma_C = 1$ if and only if the static scalar load has no first-order local comparison-volume mode.

17.7 Gauge-reduced scalar normal form

The generator formulation still permits too many apparent degrees of freedom. Some are physical, but some are only coordinate choice, anisotropic shear, or time-space mixing inappropriate to the static scalar exterior sector. The next reduction is to put the local response into normal form.

Definition 17.35 (Gauge-reduced scalar response sector). *Fix a static, spherically symmetric exterior weak-field load. A local clock-ruler response map is in the gauge-reduced scalar sector when, after quotienting coordinate relabelings, anisotropic shear, and parity-forbidden time-space mixing, its infinitesimal generator is diagonal and direction-independent:*

$$L_{\text{sc}} = \begin{pmatrix} \lambda_t & 0 \\ 0 & \lambda_s \end{pmatrix} = \frac{\theta}{2} I + \sigma H, \quad H := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Here

$$\theta := \lambda_t + \lambda_s = \text{tr } L_{\text{sc}}, \quad \sigma := \frac{\lambda_t - \lambda_s}{2}.$$

The trace θ is the comparison-cell dilation mode; the exchange coefficient σ transfers comparison scale between clock and ruler channels without changing the cell determinant.

Proposition 17.36 (Clock calibration leaves only the trace obstruction). *In the gauge-reduced scalar sector, suppose the clock response is calibrated by the redshift recovery,*

$$\sqrt{\Theta_t} = 1 + \frac{\Phi_C}{c^2} + O(c^{-4}),$$

and write the spatial response as

$$\sqrt{\Theta_s} = 1 - \gamma_C \frac{\Phi_C}{c^2} + O(c^{-4}).$$

Then the normal-form generator has

$$\lambda_t = 1, \quad \lambda_s = -\gamma_C,$$

so that

$$\theta = 1 - \gamma_C, \quad \sigma = \frac{1 + \gamma_C}{2}.$$

Therefore, once the clock sector is calibrated, the entire first-order spatial-response problem is the trace problem:

$$\gamma_C = 1 \iff \theta = 0.$$

Proof. By Definition 17.30, the loaded increments are

$$d\tau = (1 + \hat{\chi}\lambda_t)d\tau_0 + O(\hat{\chi}^2), \quad d\ell = (1 + \hat{\chi}\lambda_s)d\ell_0 + O(\hat{\chi}^2),$$

with $\hat{\chi} = \Phi_C/c^2$. Matching these expressions to the displayed clock and ruler factors gives $\lambda_t = 1$ and $\lambda_s = -\gamma_C$. The formulae for θ and σ follow from Definition 17.35. Hence $\gamma_C = 1$ iff $\theta = 0$. $\square$

Corollary 17.37 (Same-sign depletion failure in normal form). *A same-sign scalar depletion of clock and ruler increments has $\lambda_t = \lambda_s = 1$. With the clock sector calibrated, this implies*

$$\gamma_C = -1,$$

so it reproduces the redshift sign but fails the spatial weak-field sector.

Proof. If $\lambda_t = 1$ and $\lambda_s = 1$, then $-\gamma_C = 1$, hence $\gamma_C = -1$. Equivalently, the trace is $\theta = 2$, so $\gamma_C = 1 - \theta = -1$. $\square$

Proposition 17.38 (Unique scalar closure compatible with calibrated weak-field GR). *Within the gauge-reduced scalar sector with the clock coefficient fixed by redshift, the following are equivalent to first post-Newtonian order:*

$$\begin{aligned} L_{\text{sc}} \in \mathfrak{sl}(2, \mathbb{R}) &\iff L_{\text{sc}} = H \\ &\iff R_{\text{sc}} \in SL(2, \mathbb{R}) + O(c^{-4}) \\ &\iff \Theta_s = \Theta_t^{-1} + O(c^{-4}) \\ &\iff \gamma_C = 1. \end{aligned}$$

Thus the scalar weak-field closure is unique: the load must act as a trace-free clock-ruler exchange, not as a common dilation.

Proof. Clock calibration gives $\lambda_t = 1$. Tracelessness gives $\lambda_s = -1$, hence $L_{\text{sc}} = H$. This is equivalent to $\det R_{\text{sc}} = 1 + O(c^{-4})$, since the determinant changes to first order by $\text{tr } L_{\text{sc}}\hat{\chi}$. The determinant condition is $\Theta_s\Theta_t = 1 + O(c^{-4})$, which is the same as $\Theta_s = \Theta_t^{-1} + O(c^{-4})$. By Proposition 17.36, this is equivalent to $\gamma_C = 1$. $\square$

Theorem 17.39 (Scalar weak-field closure package). *Assume the static isotropic exterior weak-field sector, the metric-representation gate, universal matter coupling to the same comparison kernel, and the clock calibration supplied by the redshift recovery. After coordinate gauge, anisotropic shear, and parity-forbidden time-space mixing are removed, the following statements are equivalent to first post-Newtonian order:*

$$\begin{aligned} \text{no scalar comparison-cell dilation} &\iff \text{tr } L_{\text{sc}} = 0 \\ &\iff L_{\text{sc}} \in \mathfrak{sl}(2, \mathbb{R}) \\ &\iff \det R_{\text{sc}} = 1 + O(c^{-4}) \\ &\iff \frac{d}{d\hat{\chi}} \log \sqrt{|\det q_{\text{sc}}(\hat{\chi})|} \Big|_0 = 0 \\ &\iff \Theta_s\Theta_t = 1 + O(c^{-4}) \\ &\iff \gamma_C = 1. \end{aligned}$$

If these equivalent conditions hold, the conditional weak-field exterior recovery theorem applies. If they fail with trace θ , then

$$\gamma_C = 1 - \theta,$$

so two-sector weak-field observations measure the scalar comparison-cell dilation directly.

Proof. Clock calibration gives $\lambda_t = 1$. In the gauge-reduced scalar sector, Proposition 17.36 gives $\theta = 1 - \gamma_C$, so $\gamma_C = 1$ iff $\theta = 0$. The equivalence between vanishing trace, membership in $\mathfrak{sl}(2, \mathbb{R})$, first-order unimodularity of R_{sc} , and preservation of the normalized clock-ruler cell is the determinant identity for a first-order $GL(2)$ response. In the isotropic two-sector kernel this determinant condition is $\Theta_s \Theta_t = 1 + O(c^{-4})$. The conditional weak-field recovery then follows from Theorem 17.49. If $\theta \neq 0$, Proposition 17.28 gives the shifted coefficient $\gamma_C = 1 - \theta$. $\square$

Audit Note 17.40 (What the scalar package reduces). The spatial problem is no longer an unconstrained metric-fitting problem. In the static scalar exterior sector, after gauge and shear are removed, redshift fixes λ_t . The only remaining first-order scalar ambiguity is $\theta = \text{tr } L_{\text{sc}}$. Proving the APF substrate has no independent scalar comparison-cell dilation is therefore enough to close the leading two-sector weak-field geometry.

Gate 17.41 (Unimodular scalar load response). *Static scalar geometric recruitment acts on each local clock-ruler comparison plane by a unimodular response after coordinate gauge, anisotropic shear, and composition-dependent channels are removed:*

$$\det R_e(\hat{\chi}) = 1 + O(c^{-4}), \quad \text{equivalently} \quad L_e \in \mathfrak{sl}(2, \mathbb{R}).$$

Physical reading: scalar recruitment may redistribute comparison capacity between clock and ruler channels, but it does not create or destroy the elementary local comparison cell. A failure of this gate is not a small technical issue; it is an additional scalar trace mode in the weak-field geometry.

Proposition 17.42 (Unimodular response closes the spatial coefficient). *Assume the two-sector weak-field kernel and the unimodular scalar load response gate. Then*

$$\Theta_s \Theta_t = 1 + O(c^{-4}), \quad \gamma_C = 1.$$

Proof. The determinant of the clock-ruler response map is the normalized comparison-cell factor:

$$\det R_e = \mathcal{A}_e = \left(\frac{d\tau}{dt} \right) \left(\frac{d\ell_e}{dx_e} \right) = \sqrt{\Theta_t \Theta_{s,e}}.$$

Unimodularity gives $\mathcal{A}_e = 1 + O(c^{-4})$, hence $\Theta_t \Theta_{s,e} = 1 + O(c^{-4})$. In the isotropic sector this is $\Theta_t \Theta_s = 1 + O(c^{-4})$, and Proposition 17.7 gives $\gamma_C = 1$. $\square$

Corollary 17.43 (Equivalence of the spatial closure targets). *In the isotropic weak-field regime, the following are equivalent to first post-Newtonian order:*

$$\begin{aligned} \text{tr } L_e = 0 &\iff \det R_e = 1 + O(c^{-4}) \\ &\iff \mathcal{A}_e = 1 + O(c^{-4}) \\ &\iff \Theta_s \Theta_t = 1 + O(c^{-4}) \\ &\iff \gamma_C = 1. \end{aligned}$$

The spatial gate can therefore be pursued as any one of four equivalent problems: rule out the trace generator, prove unimodular response, prove comparison-cell conservation, or derive reciprocal clock-ruler response.

Audit Note 17.44 (Operational diagnostic for the trace mode). If weak-field tests determine an effective PPN coefficient γ_{obs} , the APF trace mode is read off as

$$\theta_e = \zeta_C = 1 - \gamma_{\text{obs}}.$$

Thus the observational role of light bending, Shapiro delay, and other two-sector probes is not merely to test GR. In the cost language, they bound or detect the first-order determinant of the

local clock-ruler response map. Redshift alone cannot do this, because it probes only the clock projection.

Proposition 17.45 (Nonzero trace is a scalar comparison mode). *Within the two-sector weak-field ansatz, a nonzero first-order trace generator $\theta_e \neq 0$ is observationally equivalent, at this order, to adding an independent scalar comparison mode that changes the spatial PPN coefficient while leaving the redshift sector fixed.*

Proof. The redshift sector fixes $\Theta_t = 1 + 2\Phi_C/c^2 + O(c^{-4})$. A nonzero trace generator changes the product $\Theta_t\Theta_s$ by

$$\Theta_t\Theta_s = 1 + 2\theta_e \frac{\Phi_C}{c^2} + O(c^{-4}),$$

which by Proposition 17.31 shifts γ_C to $1 - \theta_e$. Since Θ_t is unchanged, the new degree of freedom is invisible to clock-sector redshift but visible to two-sector spatial observables. That is exactly the weak-field role of an additional scalar comparison mode. $\square$

Remark 17.46 (Why this is a strengthening). The variational obstruction can be expressed invariantly as a generator question: is scalar load transport $GL(2)$ with nonzero trace, or $SL(2)$ to first order? This is the cleanest current statement of the spatial-response problem.

Corollary 17.47 (A sharper spatial-response target). *The weak-field spatial sector can be closed by proving the unimodular scalar load response gate Gate 17.41, or equivalently any of the targets listed in Corollary 17.43. One convenient proof chain is*

$$\begin{aligned} \text{tr } L_e = 0 &\implies \det R_e = 1 + O(c^{-4}) \\ &\implies \Theta_s\Theta_t = 1 + O(c^{-4}) \\ &\implies \gamma_C = 1 \\ &\implies \text{leading two-sector weak-field GR recovery.} \end{aligned}$$

Thus the next target is no longer the bare coefficient γ_C , but the underlying no-trace / unimodular comparison-load certificate.

Assumption 17.48 (Two-sector weak-field response). *In the exterior weak-field regime generated by a static point source, the represented comparison kernel has*

$$\Theta_t(r) = 1 + \frac{2\Phi_C(r)}{c^2} + O(c^{-4}), \quad \Theta_s(r) = 1 - 2\gamma_C \frac{\Phi_C(r)}{c^2} + O(c^{-4}).$$

The clock calculation gives the first relation conditionally. The second relation is the spatial-response target.

Theorem 17.49 (Conditional weak-field exterior recovery). *Assume the metric-representation gate, the exterior point-source profile, the Newtonian calibration $\Phi_C = -GM/r$, the proper-time dictionary, universal matter coupling to the same comparison kernel, and the spatial-response gate $\gamma_C = 1$. Then the represented exterior weak-field cost kernel is*

$$ds_C^2 = - \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 + \left(1 + \frac{2GM}{rc^2}\right) d\mathbf{x}^2 + O(c^{-4}),$$

which is the first-post-Newtonian Schwarzschild exterior metric in isotropic weak-field form. Consequently the framework conditionally recovers, at leading weak-field order, gravitational redshift, clock-height tests, Newtonian acceleration, light bending, and Shapiro time delay.

Proof. The clock-sector recovery and calibration give

$$\Theta_t(r) = 1 - \frac{2GM}{rc^2} + O(c^{-4}).$$

The spatial-response gate with $\gamma_C = 1$ gives

$$\Theta_s(r) = 1 + \frac{2GM}{rc^2} + O(c^{-4}).$$

Substitution into Definition 17.1 gives the displayed metric kernel. In standard post-Newtonian bookkeeping, this pair of coefficients is exactly the weak-field Schwarzschild exterior through the leading order controlling the listed observables. The recovery is conditional because the metric-representation gate, universal matter coupling, and either the spatial-response gate or its sharper reciprocal clock-ruler route are not yet derived from substrate primitives. $\square$

Corollary 17.50 (No-trace scalar response is sufficient for the leading weak-field package). *Under the hypotheses of Theorem 17.49, the explicit spatial-response assumption $\gamma_C = 1$ may be replaced by any one of the equivalent scalar-closure conditions in Theorem 17.39. In particular, if the gauge-reduced scalar load generator is traceless,*

$$\text{tr } L_{\text{sc}} = 0,$$

then the conditional first-post-Newtonian exterior recovery follows.

Proof. By Theorem 17.39, vanishing trace is equivalent to $\gamma_C = 1$ in the static isotropic scalar exterior sector after clock calibration. Substituting this equivalent gate for the spatial-response assumption gives exactly the hypotheses of Theorem 17.49. $\square$

Corollary 17.51 (Observable dependency split). *Within the two-sector weak-field kernel, redshift and clock-height tests are clock-sector observables, while light bending and Shapiro delay are two-sector observables. Therefore any APF claim to recover leading weak-field GR must close both the clock-response gate and the spatial-response gate.*

Observable	Needs Θ_t	Needs Θ_s	Present status
Static redshift	Yes	No	Recovered conditionally from clock response.
Clock-height tests	Yes	No	Same gate as redshift, with local calibration.
Newtonian acceleration	Yes	No at leading slow-body order	Conditional on geodesic-cost dictionary and universal matter coupling.
Light bending	Yes	Yes	Open until $\gamma_C = 1$ is derived.
Shapiro delay	Yes	Yes	Open until null comparison uses the two-sector kernel with $\gamma_C = 1$.
Perihelion precession	Yes	Yes plus next-order terms	Partly first-PN; full recovery needs nonlinear corrections.
Frame dragging	Needs h_{0i}	Yes	Deferred to rotating-source tensor sector.

Gravitational radiation	Tensor dynamics	Tensor dynamics	Deferred to dynamical self-coupled geometric substrate.
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Audit Note 17.52 (No-smuggling table for the weak-field suite).

18. From response cost geometry to tensorial comparison response

The scalar calculation used one effective load χ and one clock stiffness. The intrinsic construction now shows that a tensorial theory may carry two different quadratic objects:

$$g_{ab}^{\text{cost}} \quad \text{and} \quad q_{ab}^{\text{cmp}}.$$

The first, when it exists, is the positive metric induced by calibrated response effort. The second is the signed comparison kernel supplied by the causal/signature gate. Their relation is a bridge theorem, not an identity inherited from notation.

Definition 18.1 (Tensorial signed comparison kernel). *A tensorial signed comparison kernel is a smooth nondegenerate field q_{ab}^{cmp} whose quadratic function*

$$Q_x^{\text{cmp}}(dx) = q_{ab}^{\text{cmp}}(x) dx^a dx^b$$

represents the causal comparison structure. When the signature gate closes it has Lorentzian signature. The positive implementation effort of the same displacement is measured separately by g^{cost} or by the underlying Finsler gauge.

Definition 18.2 (Weak perturbation around unloaded comparison geometry). *In the represented weak-field regime, write*

$$q_{ab}^{\text{cmp}} = \eta_{ab} + h_{ab}, \quad |h_{ab}| \ll 1,$$

where η_{ab} is the unloaded Lorentzian comparison kernel and h_{ab} the recruited comparison distortion.

Principle 18.3 (Universal comparison-load coupling). *Matter distinctions couple to the represented comparison geometry through the burden they impose on the substrate. In the tensorial regime this burden is summarized by a stress-energy functional T^{ab} , and the leading coupling has schematic form*

$$E_{\text{rec}}[h; T] = E_{\text{geom}}[h] - \lambda \int h_{ab} T^{ab} d^4x + O(h^2 T, h T^2).$$

Universality of free fall requires the same comparison kernel to be read by all admissible matter sectors. The positive cost geometry must be compatible with this coupling through the anchor and cost-causal bridge.

Gate 18.4 (No composition dependence gate). *The equivalence-principle-like claim closes only if the recruitment load of a composite body depends on its total carried stress-energy distinction and not on microscopic composition except through already-accounted internal energy. All admissible clocks and test bodies must read the same signed comparison kernel and its calibrated cost response.*

Proposition 18.5 (Redshift sector as the h_{00} projection). *In the weak static regime, redshift fixes*

$$h_{00}(r) \simeq \frac{2\Phi_C(r)}{c^2} = -\frac{2GM}{rc^2}$$

up to the sign convention of the comparison kernel. It does not fix h_{ij} , h_{0i} , or the positive cost metric g^{cost} away from the calibrated clock direction.

Proof. Static redshift compares rest-frame time increments and therefore projects the signed comparison kernel onto its time-time component. Spatial, mixed, and independent cost-geometry components require additional probes and bridge laws. $\square$

19. Translation route to the Paper 6 field equations

The Paper 6 route to GR is not to guess the Schwarzschild metric, but to characterize the admissible local cost functional for Q_{ab} after the represented smooth Lorentzian gates close. The expected structure is:

$$E_{\text{geom}}[Q] = \int_{\mathcal{M}} \left[\text{local scalar made from } Q \text{ and its first/second derivatives} \right] d\mu_Q.$$

Paper 6 supplies this represented-geometry closure using H1–H3, A9, and Lovelock uniqueness. Papers 24/25 already discharge H1 and H2 (continuum emergence and locality of the Δ_{geo} closure) at the recruitment level, so the continuum hypotheses needing translation reduce to H3 and the A9/Lovelock step. The Paper 9 challenge is narrower: translate those residual hypotheses into finite-continuability certificates and identify their weak-field scalar face.

Gate 19.1 (Coordinate-relabeling invariance gate). *The cost assigned to a comparison-continuation configuration may not depend on arbitrary labels used for comparison sites. In the smooth represented regime, this becomes diffeomorphism-type invariance of the geometric cost functional.*

Gate 19.2 (Second-order locality gate). *If the substrate cost is local and the represented field equations are to avoid additional higher-derivative propagating comparison modes, the continuum Euler-Lagrange equations should be second order in the metric kernel. This is the gate that points toward curvature scalar functionals rather than arbitrary higher-curvature actions.*

Gate 19.3 (Bianchi/ledger gate). *The geometric side of the represented field equation must have an identically conserved ledger divergence so that local stress-energy continuation is compatible with the geometry equation. In GR this is the contracted Bianchi identity. In cost language it is the statement that relabeling comparison sites cannot create or destroy committed load.*

Definition 19.4 (Geometric Euler ledger). *Let $E_{\text{geom}}[Q]$ be a represented geometric cost functional. Its Euler ledger is the variational tensor density $\mathcal{E}^{ab}$ defined by*

$$\delta E_{\text{geom}}[Q] = \int \mathcal{E}^{ab} \delta Q_{ab} d\mu_Q + \text{boundary terms}.$$

A matter recruitment functional contributes a ledger $\mathcal{T}^{ab}$ in the same variational channel.

Proposition 19.5 (Relabeling invariance implies a conserved geometric ledger). *If $E_{\text{geom}}[Q]$ is invariant under compactly supported infinitesimal relabelings of comparison sites, then its Euler ledger satisfies a formal conservation identity*

$$\nabla_a \mathcal{E}^{ab} = 0$$

in the represented smooth metric regime, up to the boundary conventions of the functional.

Proof sketch. An infinitesimal relabeling generated by a vector field ξ^a changes the metric kernel by its Lie derivative,

$$\delta Q_{ab} = \mathcal{L}_\xi Q_{ab} = \nabla_a \xi_b + \nabla_b \xi_a.$$

Relabeling invariance gives $\delta E_{\text{geom}} = 0$ for all compactly supported ξ . Substituting the variation and integrating by parts yields

$$0 = -2 \int (\nabla_a \mathcal{E}^{ab}) \xi_b d\mu_Q.$$

Since ξ is arbitrary, $\nabla_a \mathcal{E}^{ab} = 0$. The result is a represented-regime conservation identity; translating it back to finite continuations is a separate APF gate. $\square$

Corollary 19.6 (Matter compatibility condition). *If the represented field equation has the form*

$$\mathcal{E}^{ab} = \lambda \mathcal{T}^{ab},$$

then relabeling invariance of the geometric ledger requires

$$\nabla_a \mathcal{T}^{ab} = 0$$

inside the same represented regime. In APF terms, local committed recruitment load must continue consistently with the comparison geometry that carries it.

Theorem 19.7 (Translation target: Einstein-Hilbert uniqueness, schematic). *If the represented geometric cost functional is local, smooth, coordinate-relabeling invariant, built only from the tensorial comparison kernel and its derivatives, yields second-order equations, and has a conserved ledger divergence compatible with universal stress-energy recruitment, then the leading continuum functional is of Einstein-Hilbert type,*

$$E_{\text{geom}}[Q] \propto \int (R[Q] - 2\Lambda) d\mu_Q,$$

up to boundary terms, normalization, and higher-order corrections outside the gate.

Boundary 19.8 (Status of Theorem 19.7). This theorem is not reproved here. It is the Paper 9 translation target corresponding to the Paper 6 A9/Lovelock closure. Paper 6 supplies the represented-geometry proof; Paper 9 asks which finite-continuity certificates force the same continuum hypotheses and, in the weak-field scalar sector, which certificate forces the no-trace condition $\theta = 0$.

20. Nonlinear alignment: from response law to the Paper 6 closure

The leading-order theory uses a scalar accumulated load χ . Paper 6 supplies the represented tensorial, nonlinear, self-coupled closure to the Einstein equations. The remaining Paper 9 problem is to show how the finite comparison-substrate response lifts into exactly the Paper 6 tensorial closure rather than an alternative scalar-tensor or higher-derivative geometry.

Gate 20.1 (Tensorial geometric response gate). *The scalar stiffness K_t must lift to a tensorial comparison-cost structure g_{ab} or equivalent cost kernel such that matter recruitment load changes the full comparison geometry, not only the static time-comparison sector.*

Gate 20.2 (Self-coupling gate). *The geometric recruitment load must include the cost of the geometric substrate's own recruited configuration. In GR language, gravitational field energy gravitates. In cost language, committed comparison-cost distortion contributes to the future comparison-cost kernel.*

Gate 20.3 (Einstein-equation gate). *The nonlinear geometric recruitment functional must have Euler-Lagrange equations equivalent, in the represented smooth Lorentzian regime, to*

$$G_{ab} + \Lambda g_{ab} = \frac{8\pi G}{c^4} T_{ab},$$

with G remaining either a substrate-specific input or a derived stiffness constant.

Boundary 20.4 (Primary translation theorem). The Paper 6 endpoint can be restated in Paper 9 language as follows:

If finite comparison continuations admit a complete calibrated response geometry, a physical comparison-response anchor, a smooth local Lorentzian signed comparison kernel, universal recruitment coupling, local A9-type closure, and nonlinear self-consistent response, then their represented continuum equations are the Einstein equations.

Paper 6 proves the represented-geometry closure under its hypotheses. Paper 9 has not yet proved that finite continuability forces those hypotheses. Its current hard local target is the weak-field trace theorem $\theta = 0$, the scalar shadow of the same closure.

Object	Status	Role here
Held continuation profile	I	Imported from Papers 1, 10, and 14 as the pre-record identity carrier.
Record-forming class transition	I	Imported from Paper 37; downstream of the held response geometry.
Complete represented profile carrier	I	Imported from Papers 1 and 10; Paper 5 supplies the quantum-sector completion.
Local quantum orientation	I, quantum only	Imported from Paper 5 only in its declared coherent quantum sector; not identified with clock-ruler or causal orientation.
Geometric response quotient	P H	Proved here under constant-rank regularity.
Substrate anchor	O/C	Identity-complete physical anchor still to be constructed.
Lorentzian signed comparison kernel	O/C	Requires independent cone and signed two-homogeneous gates.
No-trace weak-field source	O/C	The invariant target remains $P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$.

Audit Note 20.5 (Current cross-corpus status).

21. Input ledger and closure plan

Claim / object	Status	What remains
Continuation judgment $\Gamma \Vdash_C D \rightsquigarrow E$	P	Imported from Paper 10 as primitive grammar.
Finite operational basis	R/G+C	General theorem requires joint extension, minimum attainment, closure invariance, and positive marginal extension cost; finite atom-cover model is Engine-certified.
Operational basis to physical precision channels	O/G	Construct the faithful realization preserving joint realizability and recoverability, separating reduced states, and admitting real limits.
Record topology and second countability	P G	Hausdorffness requires state-separating real channels; second countability requires the countable grammar and topology-generation gate.
Finite smooth generation	O/G	Prove local smooth functional generation by the finite channel map.
Smooth response-factor completeness	O/G	Prove smooth two-way factorization between the complete response and a finite local APF generator before using the derivative-kernel identity.

Regular response quotient	P H	Proved locally after constant rank and smooth response-factor completeness; the physical comparison response model and rank stratification remain to be constructed.
Response-orbit gauge and distance	P	Proved intrinsically on the regular quotient; smooth strongly convex Finsler structure requires additional norm regularity.
Linear Paper 2 specialization	P	Proved as the flat quotient branch with optimal response gains.
Physical-ledger calibration	C/P	Comparison theorem proved; actual ledger-rate bounds and response-class completeness remain physical inputs.
Residual energy–distance relation	P SRR	Proved locally when the residual Hessian is positive definite.
Physical comparison-substrate anchor	O/C	Construct s and $A = q \circ Ds$, type the null kernel and provenance, and prove direct-germ calibration.
Quadratic positive cost metric	P Q	Proved from the parallelogram certificate; whether the substrate passes it is open.
Signed Lorentz–Finsler comparison structure	O/C	Derive the causal cone, orientation, signed two-homogeneous function, and Lorentzian Hessian.
Dimension lock-in	O	Requires an anchor-aware mode-count theorem, not merely register count.
Point-source $1/r$ profile	C/P	Proved from an assumed isotropic quadratic exterior kernel.
Clock response and red-shift	R	Conditional recovery under load law, calibration, and proper-time dictionary.
Trace–volume identity	P/C	Proved in the represented clock–ruler comparison plane.
Gauge-fixed response Hessian	O/C	Derive $\mathcal{K}$ and $\mathcal{J}$ from the recruitment master equation after gauge removal.
No-trace operator $P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$	O/C	Decisive local theorem; source-selection and fixed-volume are the principal routes.
Spatial response $\gamma_C = 1$	C/O	Follows from the no-trace operator through the trace–volume theorem.
Conditional weak-field exterior	C/R	Follows from clock calibration, universal coupling, and no trace.
Full Einstein equations	I/C	External Paper 6 endpoint; residual finite-continuity translation remains open.

22. Formal theorem tree for closure

ID	Claim	Status	Dependency
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G9.1	A finite complete operational basis exists.	R G/C	General joint-extension package; exact atom-cover Engine witness.
G9.2	The operational basis is realized as finite separating precision channels.	O/G	Faithful operational-to-channel realization.
G9.3	The record topology is Hausdorff and second countable.	P G	State separation; countable grammar; topology-generation gate.
G9.4	Finite smooth generation gives a subcartesian differential space.	P G	Finite smooth-generation gate.
G9.5	Complete local continuation flows generate smooth orbits.	I/P	Subcartesian orbit theorem.
G9.6	Smooth response-factor completeness gives $\ker D\beta = N^{\text{ctx}}$.	P G	Two-way smooth factorization gate.
G9.7	The process norm defines an intrinsic response-orbit quotient gauge and length.	P	Theorems 10.55 and 10.57.
G9.8	A physical anchor pulls response-orbit length back to site geometry.	P G	Gate 10.58 and Theorem 10.59.
G9.9	Calibrated physical path cost is bi-Lipschitz to response distance.	P H	Theorem 9.2.
G9.10	Smooth residual energy is one half squared local distance.	P SRR	Theorem 9.4.
G9.11	Parallelogram quadraticity gives a positive cost metric.	P Q	Proposition 10.60.
G9.12	Positive pullback geometry cannot supply Lorentzian signature.	P	Proposition 11.1.
G9.13	A signed Lorentz–Finsler comparison structure represents the causal cone.	O/C	Gate 11.4.
G9.14	Isotropic quadratic exterior recruitment gives $\chi \sim 1/r$.	C/P	Proposition 12.4.
G9.15	Calibrated clock response conditionally recovers redshift.	R	Response law and proper-time dictionary.
G9.16	Generator trace equals comparison-volume variation.	P/C	Theorem 17.33.
G9.17	Linear loaded equilibrium is $h_\chi = \chi \mathcal{K}^{-1} \mathcal{J} + O(\chi^2)$.	P H	Theorem 17.23.
G9.18	The no-trace gate is $P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0$.	O/C	Source-selection or constrained-volume theorem.

G9.19	No trace gives $\gamma_C = 1$ and the first-PN exterior.	C/R	Corollary 17.34 and Theorem 17.49.
G9.20	The represented nonlinear closure is Einstein gravity.	1/C	External Paper 6 H1–H3/A9/Lovelock endpoint.

23. Minimal publishable core

A stable geometry-centered version of Paper 9 should contain:

1. the regular response quotient and pullback-gauge theorem;
2. the intrinsic path-distance and linear Paper 2 recovery theorem;
3. the physical-ledger calibration theorem and the length–energy fork;
4. the intrinsic response-orbit gauge, the physical comparison-anchor gate and conditional pullback theorem, and the explicit separation of positive cost from Lorentzian comparison geometry;
5. the trace–volume theorem in the loaded clock–ruler plane;
6. the no-trace operator reduction $P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$;
7. the existing conditional redshift and first-PN recovery package, with Paper 6 imports kept explicit.

This is a stronger mathematical identity for the paper than “cost accounting plus a weak-field bridge.” It makes Paper 9 the intrinsic differential geometry of admissibility response while preserving the precise gravitational target.

Audit Note 23.1 (Defer connections and curvature). A connection is not canonical on a general nonsmooth or non-strongly-convex Finsler structure, and rank-changing response spaces may not be manifolds. Curvature should therefore be introduced only after the regular quotient, smoothness, strong convexity or quadraticity, anchor, and causal signature gates have closed. The trace–volume theorem does not require that later machinery.

24. Research tasks for the next draft

1. **Construct the physical response manifold.** Identify the actual comparison-response coordinates in the recruitment formalism, compute the rank of the response map, and classify rank-changing strata.
2. **Calibrate the ledger rate.** Prove bounds $aF_{\text{resp}} \leq j \leq bF_{\text{resp}}$ on a complete declared response class, or exhibit fixed setup, threshold, or hysteresis terms that force a non-length branch.
3. **Derive the comparison-response anchor.** Specify how an infinitesimal site displacement recruits a quotient response direction and prove agreement with the direct comparison germ.
4. **Test quadraticity rather than assume it.** Measure or derive parallelogram defects. Retain the Finsler branch if weighted or polyhedral cost survives.
5. **Build the causal comparison function.** Derive the finite-coordination cone, irreversible orientation, and a signed two-homogeneous Q^{cmp} with Lorentzian Hessian.

6. **Compute the loaded response operator.** Expand the Papers 24/25 recruitment master equation in the comparison-kernel sector to identify $\mathcal{K}$ and $\mathcal{J}$ after gauge removal.
7. **Prove the no-trace operator.** Pursue either $P_{\text{tr}}\mathcal{J} = 0$ plus Hessian block-diagonality, or a primitive fixed-volume constraint. The target is $P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$, not merely a fitted γ_C .
8. **Interpret the weight-one program microscopically.** Treat its capacity accounting as a candidate mechanism for source-selection, not as a substitute for the general response theorem.
9. **Freeze Paper 2.** Once these theorems stabilize, Paper 2 should import the finite linear specialization and its physical-ledger bridge rather than develop a parallel geometry section.

A. Candidate microscopic realization of the no-trace gate: the weight-one program

The trace gate can be priced in capacity terms, and the price splits. Write the static response as a direct clock coupling plus a spread deposit of weight w , distributed η -democratically over the comparison cell with denominator $n - 1$, where n is the spatial dimension. We adopt this decomposition and this spread shape; neither is derived, and the ledger below carries them as S0 and S1. The spatial coefficient is then a one-parameter family,

$$\gamma_C(w) = \frac{w}{(n - 1) - w},$$

and general relativity is the weight-one point in every dimension: $\gamma_C(1) = 1/(n - 2) = \gamma_{\text{GR}}(n)$. The closure candidates for the response sit at fork coordinates on this curve — the plane rule at $w = 1$, the four-cell rule at $w = 1/2$, the per-channel rule at $w = 0$, the pooled rule at no finite weight — so the gate question becomes a question about one scalar weight (`check_T_gammaC_carrier_fork`, `gamma_c_carrier_program.py`).

The weight has an exact accounting reading. Let D be the cell-sum of a deposit pattern over the $n + 1$ channel accounts of the comparison cell. Then $w = D(\text{spread deposit})/D(\text{commitment})$, identically in n and w : the η -diagonal sums to $n - 1$ over the cell, so the denominator is not an independent import but the unique normalization under which the weight reads as a fraction of the committed amount. The same cell-sum annihilates every clock-ruler exchange pattern, so the redshift sector carries zero net amount and cannot be over-billed by this accounting. And $w \neq 1$ is exactly one identically conserved scalar improvement mode of strength proportional to $1 - w$ — the independent scalar comparison mode of Proposition 17.45, met in ledger coordinates (`check_T_weight_one_reduction`, `gamma_c_carrier_program.py`). In this coordinate the no-trace gate is a conservation statement: the substrate's cell-side deposit matches the load's committed amount.

Matching splits into two halves with different prices. The no-source half needs the minimal trace-currency premise TCP-min — the response deposit is denominated in committed capacity at a channel-uniform rate — a named conjecture, not a theorem. Under it the deposit must be billed to the load's own committed line: no census sector furnishes a second local payer, and a line cannot be overdrawn. Hence $D(\text{deposit}) \leq D(\text{commitment})$, that is $w \leq 1$ and $\gamma_C \leq 1/(n - 2)$; at $n = 3$,

$$\gamma_C \leq 1 \quad \text{at} \quad [P_{\text{structural}} \mid \text{S0/S1} + \text{TCP-min} + \text{H5-W} + \text{census legs} \\ + \text{the Paper 9/Paper 10 base (P5, B7 incl. adopted } C_M)].$$

The bound is citable only with that ledger attached. H5-W is census transport re-fenced to net-deposit objects — a named premise, separate from the rest and still standing on its own. P5 is this paper’s scalar package; B7 is Paper 10’s held-record capacity floor together with the adopted identification of the maintenance cost C_M with committed capacity. The no-sink half costs more. It needs TCP-strong, the re-expression-in-full clause: the static response is the same committed amount read at the cell, not less — the one-sided clause $D(\text{deposit}) \geq D(\text{commitment})$. Nothing cheaper closes it, because an absence of deposit separates no continuation profile and carries no bill. Composed with the bound, TCP-strong gives $w = 1$, hence $\gamma_C = 1/(n - 2)$ in every n and $\gamma_C = 1$ at $n = 3$ — still conditional on the full ledger. Neither trace-currency clause is derived.

The TCP-strong gap has a precise location in Paper 10. Testedness there is binary: a distinction-expression either separates a tested continuation profile or preserves every one. The amount-graded refinement that would close the gap — billing a partial expression for its missing fraction — does not derive from the calculus. The obstruction is an encoding wall: the calculus grades the tested-atom count, the count is representation data, and floor-coherent signatures hold the same record at grades $\varepsilon^* \cdot \{1, \log_2 N, N\}$ depending on how it is atomized, with ε^* the distinction floor (`check_L_amount_graded_testedness_encoding_no_go`, `amount_graded_testedness.py`). What survives is an anatomy: over the adopted linear frame the discharge splits into a distributed-encoding existence premise and a par billing rate, with the dimension audit selecting par within the linear family — a selection, not a discharge. Whether the substrate holds a static response profile as one compressed register or as one tested unit per resolution rung is a record-structure question about the substrate, and it is the single genuine reopening door.

The door has a shape. In the coherent shared-read model, the only class of activation law that moves the encoding argmin toward the distributed register is a strictly superadditive read-contention law; linear and subadditive laws leave the compressed register in place (`check_T_fagt2_encoding_argmin_pressure_and_read_channel`, `amount_graded_testedness.py`). And the banked corpus prices every compatible-sharing configuration on the shut side of that gate. The derived joint-cost identity $\Delta = \varepsilon^*(J - R)$, iterated over M compatible readers of one shared record channel, prices the sharing at $\Delta = -(M - 1)\varepsilon^*$ — a strictly deepening discount; the banked activation floor is linear and saves nothing; and the recruitment corpus is reader-blind by census, with no demand-multiplicity variable anywhere in its banked signatures. The one sign-positive source in the calculus is irreducible jointness, $J > R$ — the occupancy feature. So the residual question is an occupancy fork. Either the concurrent-activation bottleneck sits at sub-record granularity — finiteness supplies the superadditive shape of such a cap for free, but the granularity is a premise finiteness does not force — or concurrent readouts of the static record are irreducibly joint at atom granularity: occupancy at the read interface, in the current Paper 5 sense of a non-Boolean held-profile sector with record-incompatible co-available continuations, per-interface and read off the world rather than derived, and not adopted here. The bill for the second horn is stated plainly: it needs the static-field read interface to present record-incompatible co-available families, and a classical broadcast readout of one static record is the separable configuration par excellence, so the natural typing sits on the adverse side (`check_T_contention_law_granularity_occupancy_fork`, `amount_graded_testedness.py`).

The bound meets observation signed. Cassini (Bertotti–Jess–Tortora 2003) gives $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$, and the central value sits 0.91σ above the bound: the bound survives at under one sigma and is under live one-sided tension, not merely saturated. A confirmed $\gamma_{\text{obs}} > 1$ at $\geq 3\sigma$ falsifies the composite (S0/S1 + TCP-min + H5-W + census + P5/B7 base). A factor-three precision improvement at the same central value puts the composite under roughly 3σ pressure. A confirmed

$\gamma_{\text{obs}} < 1$ at $\geq 3\sigma$ refutes only the no-sink clause and leaves the bound standing. The two halves are separately falsifiable, which the unsplit gate was not.

The decisive target remains the no-trace/unimodular certificate of Gate 17.41: by Proposition 17.42 and Corollary 17.43 it closes the leading spatial coefficient, and by Propositions 17.28 and 17.31 any trace source is exactly the PPN- γ obstruction. What the reduction contributes is the exact residue: over the adopted spread shape, the gate's underived content is two currency clauses, and the cheaper one already buys a falsifiable one-sided bound.

B. Source-of-record redshift calculation

This appendix records the compact calculation that motivates the first recovery theorem. Let a static point mass carry an effective accumulated recruitment load $\chi_{\text{rec}}(r)$. Suppose the exterior linear-response law is

$$K_t(r) = K_{t0}(1 - \alpha\chi_{\text{rec}}(r)),$$

and suppose the point-source equilibrium plus physical calibration gives

$$\alpha\chi_{\text{rec}}(r) = \frac{2GM}{rc^2}.$$

Then

$$K_t(r) = K_{t0} \left(1 - \frac{2GM}{rc^2} \right).$$

Using the represented proper-time dictionary,

$$d\tau(r) = \sqrt{K_t(r)/K_{t0}} dt,$$

one obtains

$$d\tau(r) = \sqrt{1 - \frac{2GM}{rc^2}} dt.$$

For two static observers at $r_B < r_A$, equal coordinate periods correspond to different local proper periods, hence

$$\frac{\omega_A}{\omega_B} = \frac{d\tau(r_B)}{d\tau(r_A)} = \sqrt{\frac{1 - 2GM/r_B c^2}{1 - 2GM/r_A c^2}}.$$

Expanding to first order gives

$$\frac{\omega_A}{\omega_B} \approx 1 - \frac{GM}{c^2} \left(\frac{1}{r_B} - \frac{1}{r_A} \right).$$

This is the static gravitational redshift form. The calculation is a conditional recovery because the response law, calibration, and proper-time dictionary remain open gates.

C. Minimal next-lemma inventory

The next version should prove, weaken, or abandon the following lemmas.

1. **Regular-quotient lemma.** The comparison-response map supplied by the recruitment formalism has constant rank on the weak-field branch, or else admits a controlled stratification.
2. **Ledger-calibration lemma.** The physical interaction rate is uniformly comparable with the pullback process gauge on the complete response class.

3. **Anchor lemma.** The direct comparison germ equals the anchored response germ for all locally admissible comparison directions.
4. **Quadraticity-or-Finsler lemma.** Determine whether the local gauge passes the parallelogram certificate; if not, retain the exact Finsler geometry rather than fitting a metric.
5. **Causal-cone lemma.** Finite coordination and irreversible record order produce a smooth oriented cone and signed comparison function with Lorentzian Hessian.
6. **Response-Hessian lemma.** Loaded-equilibrium expansion yields a gauge-fixed invertible $\mathcal{K}$ and source $\mathcal{J}$ in the comparison-kernel sector.
7. **No-trace source lemma.** Prove $P_{\text{tr}}\mathcal{J} = 0$ and $P_{\text{tr}}\mathcal{K}P_{\text{TF}} = 0$, or derive an admissibility constraint enforcing $\text{tr}_{q_0} h = 0$.
8. **Self-coupling lemma.** The cost of geometric distortion contributes to later recruitment load and lifts the weak-field response into the Paper 6 nonlinear closure.
9. **Ledger-divergence lemma.** Relabeling comparison sites induces an identically conserved geometric ledger compatible with matter continuation.

D. Conclusion

Paper 9 is now organized around a precise geometric claim. Continuation is primitive, while geometry is represented continuation. A structurally aligned held continuation profile is first represented in a complete carrier and process algebra. A finite response model then loses its redundant implementation coordinates by quotienting response fibers. The represented process norm then pulls back to a positive tangent gauge, whose path integral gives intrinsic response distance. In a linear chart this geometry is flat and reproduces the quotient norm, optimal upper gain, positive minimum gain, and compatible bi-Lipschitz comparison already established in Paper 2. A separately calibrated physical ledger may equal that distance, remain only comparable with it, or fail to be a path metric because of setup, threshold, indivisibility, saturation, or hysteresis.

The same architecture resolves the homogeneity ambiguity. One-homogeneous path ledgers generate length. Smooth residual penalties generate local quadratic energy, and their correct geometric form is one half squared distance. Neither branch should be renamed into the other. A conditional operational basis, followed by a faithful physical-channel realization, countable topology generation, and finite smooth generation, yields the record differential-space branch on which complete continuation flows form immersed smooth orbits. Smooth response-factor completeness then licenses the derivative-kernel identity and the intrinsic response-orbit quotient gauge. A physical comparison-substrate anchor is not obtained merely by naming that quotient: it remains a separate site-map and tangent-compatibility gate. Under quadraticity the intrinsic gauge yields a positive cost metric, while a separate causal cone and signed two-homogeneous structure are required for Lorentzian comparison geometry. This blocks the shortcut $g = T^*T$ as a derivation of spacetime signature while preserving the valid conditional pullback construction for positive cost.

The gravitational application becomes sharper rather than weaker. Recruitment load deforms the represented clock–ruler comparison kernel. The generator trace is exactly the first variation of the local comparison-volume element, and clock calibration turns that scalar volume mode into the PPN

obstruction $1 - \gamma_C$. In the gauge-fixed loaded-equilibrium problem the entire no-trace question is

$$P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0.$$

This equation separates two routes: show the scalar source is trace-free and the Hessian does not mix trace sectors, or derive a primitive fixed-volume admissibility constraint. The weight-one capacity program may supply a microscopic source-selection mechanism, but the invariant operator equation is the theorem it must discharge.

Paper 6 remains the represented smooth source of record for the Einstein closure. Paper 9 supplies the intrinsic response geometry underneath it: quotient, response-orbit gauge, path distance, ledger-calibration theorem, physical-anchor gate and conditional pullback, causal-signature gate, recruitment deformation, and trace-volume response. That is the coherent role of the paper and the correct home of the interaction-geometry program.

Paper 9 establishes a dependency-explicit route from finite operational comparison content to record topology, Sikorski differential structure, subcartesian regular loci, smooth continuation orbits, and positive response-orbit geometry. The general finite-basis theorem is conditional on a named joint-extension package and is machine-witnessed exactly on the finite atom-cover model. Physical precision-channel generation, finite smooth generation, smooth response-factor completeness, the physical substrate anchor, Lorentzian signature, the physical no-trace source law, and the nonlinear Einstein endpoint remain separately typed gates. The weak-field trace reduction is therefore a conditional bridge, not an unconditional rederivation of general relativity.

Compressed result.

Held continuation profiles supply the pre-record identity carrier. Response-null quotienting of their represented process realizations produces intrinsic process-induced geometry. Physical cost equals that geometry only after ledger calibration. The comparison substrate inherits it only through an anchor. Quadraticity yields a positive cost metric; causal signature requires a separate signed gate. Under recruitment load, the scalar weak-field obstruction is the local comparison-volume mode

$$\text{tr } L = \frac{d}{d\chi} \log \sqrt{|\det q_\chi|} \Big|_0 = 1 - \gamma_C,$$

and its decisive microscopic target is $P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$.